AI-BASED EMERGENCY CALL AND SMS TRIAGE SYSTEM FOR DISASTER RESPONSE IN SRI LANKAN LOW-RESOURCE LOCAL LANGUAGES

Individual component: Affect- and Uncertainty-Aware Adaptive Multi-Agent Clarification for Multilingual Emergency Calls

INDIVIDUAL THESIS

Edirisinghe E.A.K.

(IT22129758)

Group ID: R26-DS-011

Dissertation submitted in partial fulfilment of the requirements for the degree of **BSc (Hons) in Information Technology**

Specialisation: Data Science

Department of Computing

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

I declare that this is my own work and this thesis does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

I hereby grant to the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute my thesis, in whole or in part, in print, electronic or other medium. I retain the right to use this content in whole or in part in future works.

I have disclosed all use of generative AI and other automated tools in accordance with the module policy, as set out in Appendix B. I remain responsible for the accuracy, originality, citations, code, data, analyses and conclusions in this thesis.

Student name: Edirisinghe E.A. K. Student ID: IT22129758

Signature:  Date: _____10/09/2026_____

The above candidate has carried out research for the bachelor's degree thesis under our supervision.

Supervisor: Dr. Prasanna Sumathipala

Signature: _____ Date: _____

Co-supervisor: Ms. Malithi Nawarathne

Signature: _____ Date: _____

ABSTRACT

Sri Lankan emergency services receive calls in Sinhala, Tamil, English, and code-switched speech, often from people who are distressed and provide incomplete or uncertain information. In such situations, an automated system must decide what information is missing and what question should be asked next. Many existing approaches rely on fixed question templates, which can lead to repetitive clarification, weak handling of spoken numerical information, and poor management of social or off-topic calls. This research addresses these limitations by developing an affect- and uncertainty-aware adaptive multi-agent clarification component for multilingual emergency calls. The component uses dialogue state, protocol readiness, caller affect, and uncertainty to select the next clarification action through an explicit and auditable policy. The proposed solution extends a lexicon-based numeral composer to recognise hundreds, introduces a structured address parser for house number, road, and town information, applies field-specific clarification limits, and generates context-aware questions using information already provided by the caller. A finite off-topic redirect mechanism with an emergency veto was also implemented to prevent genuine emergency calls from being incorrectly dismissed. The system was evaluated using curated gold datasets, regression tests, and 502 call files compared with a fixed-template baseline. The off-topic router achieved90.4% accuracy, with no false information responses among 30 social off-topic cases. Numeral composition achieved 19/19 correct results, structured address extraction achieved 10/10, and triage classification achieved 18/20. Question diversity improved from 0.040 to 0.222, while NOT_READY_CRITICAL cases decreased from 91.4% to 74.1%. Overall, the component provides a more targeted, transparent, and testable clarification mechanism for low-resource multilingual emergency-call triage.

Keywords: emergency call triage, dialogue management, multilingual clarification, low-resource languages, off-topic control

ACKNOWLEDGEMENTS

I would like to sincerely thank my supervisor, Dr. Prasanna Sumathipala, for his guidance, valuable feedback, and continued support throughout this research. His advice helped me improve both the academic and practical aspects of my individual component. I am also very grateful to my co-supervisor, Ms. Malithi Nawarathne, for her helpful suggestions, technical guidance, and encouragement during the design, implementation, and evaluation stages of the project.

I would also like to express my appreciation to the Disaster Management Centre (DMC), Sri Lanka, for the valuable support, practical knowledge, and insights provided to the research team. Their contribution helped us better understand the real-world context of disaster-related emergency communication and made the proposed system more relevant to practical needs.

My thanks also go to my research group members for their contributions to the shared speech recognition, entity recognition, relevance analysis, and text-to-speech components, and for their support during system integration and live-panel testing. I am also grateful to the academic staff of the Department of Computing, Sri Lanka Institute of Information Technology, for the guidance, resources, and academic environment provided throughout this research.

Finally, I would like to thank my family and friends for their patience, encouragement, and continuous support during my research journey.

TABLE OF CONTENTS

1 INTRODUCTION 1

1.1 Overall project context 1

1.2 Individual component context 2

1.3 Problem statement 3

1.4 Research gap 4

1.5 Aim and objectives 4

1.6 Research questions or hypotheses 5

1.7 Scope, limitations and assumptions 5

1.8 Individual contribution 6

1.9 Thesis structure 6

2 LITERATURE REVIEW AND RELATED WORK 7

2.1 Review method and scope 7

2.2 Domain and theoretical foundations 7

2.3 Related research 8

2.4 Existing systems, products and techniques 10

2.5 Gap synthesis and conceptual framework 10

3 RESEARCH METHODOLOGY 12

3.1 Research design 12

3.2 Data, participants and study materials 13

3.3 Experimental or study procedure 14

3.4 Measures and evaluation metrics 15

3.5 Data analysis 16

3.6 Ethics, privacy, security and responsible AI 16

3.7 Reproducibility and research artefacts 16

4 PRODUCT AND SOLUTION DESIGN 17
4.1 Requirements 17
4.2 Component architecture within the complete solution 18
4.3 Design alternatives and decisions 19
4.4 Detailed design 19
4.4.1 Dialogue state and protocol readiness 20
4.4.2 Numeral composition 20
4.4.3 Structured address 21
4.4.4 Clarification policy with affect and uncertainty 22
4.4.5 Off-topic control 22
4.4.6 Controlled question templates 23
4.4.7 Dialogue state object 24
4.4.8 Risk bands and policy routes 24
4.5 Standards and professional practice 25
5 IMPLEMENTATION, INTEGRATION AND TESTING 26
5.1 Development environment and technology stack 26
5.2 Component implementation 26
5.3 Integration with the other components 27
5.4 Verification and software testing 28
5.5 Deployment and operation 28
6 RESULTS AND VALIDATION 29
6.1 Research results 29
6.1.1 Off-topic routing 29
6.1.2 Clarification comparison over the call files 31
6.1.3 Dispatch readiness 31
6.1.4 Numerals, structured address and triage 32

6.1.5 Structured address parser output on the gold set 32
6.1.6 Upstream entity and relevance models 33
6.2 Product and technical results 33
6.3 User or stakeholder validation 34
6.4 Integration results 34
6.5 Summary of findings 34
7 DISCUSSION 35
7.1 Interpretation of findings 35
7.2 Comparison with literature baselines and existing solutions 36
7.3 Research contribution 37
7.4 Product contribution 38
7.5 Limitations and threats to validity 38
7.6 Implications 39
8 COMMERCIALISATION AND SUSTAINABILITY 40
8.1 Value proposition and target users 40
8.2 Market and competitive position 40
8.3 Business or sustainability model 41
8.4 Scalability, intellectual property and compliance 41
8.5 Contribution to the Sustainable Development Goals 42
9 CONCLUSION AND FUTURE DIRECTIONS 43
9.1 Conclusion 43
9.2 Achievement of objectives 44
9.3 Individual research and product contributions 44
9.4 Future directions 44
REFERENCES 46
GLOSSARY 48

APPENDICES 49
Appendix A: Evaluation datasets and reproduction commands 49
9.4.1 Error analysis 49
9.4.2 Worked example of one call 52
9.4.3 Ask-limit simulation and question variants 52
9.4.4 Ablation study 53
9.4.5 Parameter sensitivity and calibration 55
Appendix B: Academic integrity and AI use disclosure 58

LIST OF FIGURES

Figure 2.1: Conceptual framework: four elements feeding one clarification decision 11
Figure 3.1: Off-topic gold set by bucket 14
Figure 4.1: Pipeline architecture with the individual component highlighted 18
Figure 4.2: Composition of Sinhala number words into values 21
Figure 4.3: Worked example of structured address extraction and status 21
Figure 4.4: Decision flow of the clarification policy 22
Figure 4.5: Finite redirect ladder for off-topic callers 23
Figure 6.1: Confusion matrix of the five-way route (gold rows, predicted columns) 30
Figure 6.2: Question diversity and dispatch readiness, baseline and proposed (502 call files) 31
Figure A.1: Off-topic routing accuracy by bucket 50
Figure A.2: Off-topic routing errors grouped by cause 51
Figure A.3: Off-topic ablation: route accuracy by condition 54
Figure A.4: House-number evidence window sensitivity 56
Figure A5: Relevance classifier reliability: DEV and TEST 57

LIST OF TABLES

Table 2.4.1: Comparison of approaches to clarification and call-taking 10
Table 3.2.1: Evaluation materials used in this component 13
Table 3.4.1: Definitions of the evaluation measures 15
Table 4.1.1: Functional requirements of the clarification component 17
Table 4.1.2: Non-functional requirements of the clarification component 18
Table 4.3.1: Design alternatives and the adopted decision 19
Table 4.4.1: Protocol slots, source fields and ask order 20
Table 4.4.2: Example of a refine question for a missing road, by attempt 23
Table 4.4.3: Principal fields of the dialogue state object 24
Table 4.4.4: Risk bands and the policy routes they select 24
Table 5.1.1: Technology stack 26
Table 5.4.1: Test suites and status 28
Table 6.1.1: Off-topic routing accuracy by difficulty stratum 29
Table 6.1.2: Off-topic routing accuracy by gold bucket 30
Table 6.1.3: Clarification outputs: fixed-template baseline and proposed policy 31
Table 6.1.4: Results of the numeral, address and triage components 32
Table 6.1.5: Structured address parser output on the gold set 32
Table 6.1.6: Upstream model results (group components) 33
Table 7.5.1: Limitations and threats to validity 38
Table 8.5.1: Contribution of the component to the Sustainable Development Goals 42
Table 9.2.1: Achievement of the objectives in Section 1.5 44
Table 9.4.1: Off-topic routing errors (14 of 146) 50
Table A.2: Triage gold items that were banded incorrectly 51
Table A.3: Worked example: location status and next question across two turns 52
Table A.4: Next target by number of earlier location asks 52
Table A.5: Address question variants by language and missing component 53
Table A.6: Ablation of the off-topic router (146 items) 54
Table A.7: House-number evidence window sweep 56
Table A.8: Relevance classifier reliability bins 57

LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
ASK_*	Clarification strategy codes (ASK_DIRECT, ASK_REFINE, ASK_SOFT)
BiLSTM-CRF	Bidirectional Long Short-Term Memory with a Conditional Random Field layer
CAP	Common Alerting Protocol
CRF	Conditional Random Field
DMC	Disaster Management Centre (Sri Lanka)
EENA	European Emergency Number Association
FSM	Finite-State Machine
LLM	Large Language Model
NENA	National Emergency Number Association
NER	Named Entity Recognition
NOT_READY_CRITICAL	Dispatch readiness level with critical information missing
SDG	Sustainable Development Goal
SMS	Short Message Service
STT	Speech-to-Text
TTS	Text-to-Speech
WER / CER	Word Error Rate / Character Error Rate
XLM-R	Cross-lingual Language Model, RoBERTa variant

1 INTRODUCTION

1.1 Overall project context

Emergency communication is a problem of information, as well as of speech, in an urgent situation. The caller might have provided all of the crucial details in the wrong order, switched languages on the fly, repeated things owing to nervousness, or did not supply a specific location due to the assumption of the operator knowing about it. This means that emergency communication is also a problem of managing a sequence for an automated assistant, who needs to keep in mind what is known, separate facts from assumptions, and request information only when necessary.

The demand that is set out in the context of the group solution gives rise to significant dependency between recognition in the early stages and dialogue control in subsequent ones. The outputs of speech-to-text conversion and entity recognition constitute the potential evidence, but this does not mean that the decision on the readiness of an emergency call to be dispatched is made automatically. Dialogue policy is needed to transform partial evidence into safe action, and the individual module functions at the level of decision making.

Emergency lines are the last point of contact between the public and authorities in cases of floods, landslides, fires, road accidents, and other emergencies. In Sri Lanka, these lines get phone calls from individuals in Sinhala, Tamil, English, and even in combinations of these languages, and they are usually frightened people speaking in short sentences, sometimes not knowing the name of the road where they are located. Call takers have to gather basic information before any rescue teams are sent out. Internationally, there are categories of emergency call taking like location, incident, number of people, and life safety issues [16].

The research work, the current thesis being a part of, is called AI-based emergency call and SMS triage system for disaster response in Sri Lankan low-resource local languages. It involves developing an automated pipeline that would transcribe a call, recognize the protocol information provided, determine the follow-up questions to be asked and ask those questions out loud. The pipeline is used among the four modules

implemented by the team. The present thesis describes the particular module that oversees dialogue management.

1.2 Individual component context

The component itself has been kept narrow in its purpose, but safety-critical in its importance. The component does not seek to override the policies regarding emergency dispatch, incident diagnosis, or independent authorization of resource use. Instead, its purpose is to keep an organized account of the conversation and provide a limited clarification task that will help the human operator. This division of responsibilities helps in making the component more traceable because the reason for posing a question can be connected to a protocol slot deficiency.

The term multi-agent in this thesis means separate software agents that have their own responsibilities as opposed to separate autonomous conversing agents. The dialogue state tracker will provide the evidence and the history of the conversation, while the policy agent makes the decision on what action to take. Finally, the realizer encodes the action into controlled multi-lingual language.

This module is named Affect- and Uncertainty-aware Adaptive Multi-agent Clarification for Multilingual Emergency Calls. Its inputs include the information obtained from the call in the form of a fusion of the call, i.e., the transcribed call, the spans obtained by the recognition modules, relevance label and dialogue act labels, protocol readiness information, and any distress signs. The outputs of this module include the clarification request (i.e., direct questioning, asking for clarification, confirmation, mitigation, handling of irrelevant speaker, etc.) along with the question.

However, the component is deliberately not a large language model. Its decisions are guided by an explicit state machine, whose inputs and transitions can be analyzed, tested, and justified to the supervisor or jury. Such design follows both from the necessity of ensuring safety in case of emergency calls – wrong redirections or silence may be very dangerous – and from the low resource environment itself.

1.3 Problem statement

Not only is the impact of such issues not limited to the quality of conversation; repetition may cause frustration for the caller, inability to extract the address may cause delays in getting ready to reach the location, and unregulated processing of off-topic calls may eat up the line. On the other hand, the application of an aggressive off-topic filter may be dangerous if it transfers the caller who does not provide full proof of his/her emergency.

The task can then be formulated as choosing an action a_t out of a bounded set of actions depending on the dialogue state s_t. The dialogue state consists of known spans, protocol-slot status, previous asks, missing fields, relevance, dialogue act, and caller's distress and uncertainty. An action must meet all protocol ordering and safety constraints prior to formulating its wording. This framework will be utilized consistently in the following Chapters 3-6.

The issues associated with the problem being tackled in this thesis can be identified into three categories. The first issue is that the question sequence in the call taker is policy, but the current system used by the group for the pipeline chose questions from predefined templates according to fields and thus repeated the questions. The second issue is that the location spoken in the Sinhala language, particularly the house number in the form of words, is extracted in such a way that the call ends up at a poor level of location despite having the complete address.

This is also supported by an initial observation made from the group's call files. In the call files where there are missing fields totaling 502, 91.4% of these calls were classified as NOT_READY_CRITICAL using a null check baseline, while the generated questions using the static template had a very small vocabulary (Question diversity = 0.040). Review of calls from the live panel revealed one address not captured by the system, the house number stated verbally, and a social shop opening statement regarded as an emergency call.

1.4 Research gap

A further gap is evaluation at the boundary between language processing and dialogue policy. It is entirely feasible that a classifier can work excellently on the classification side but ask the user repetitive and insensitive questions. Conversely, a good policy design may be hampered by extraction flaws at a higher level. In light of this, this thesis draws a line between component testing and pipeline comparisons.

Therefore, the contribution is described as an open and low resource system for managing dialogues, not the creation of an automated system for emergencies. The research explores whether it is possible to enhance the next question's quality through explicit state, uncertainty handling, structuring of location information, and off-topic management that remains open and replicable.

Existing spoken dialogue research formalises dialogue management as a partially observable decision process [8]-[10], and emergency-specific work focuses either on call classification or on speech recognition for emergency vocabulary. Few studies combine (a) protocol-aware slot completeness, (b) affect and uncertainty signals from the caller, (c) finite handling of off-topic callers, and (d) lexical handling of Sinhala number words, in a transparent policy that can be evaluated on a multilingual gold set. The gap addressed here is the absence of a testable, protocol-grounded clarification policy for low-resource emergency speech.

1.5 Aim and objectives

This task entails the development of an effect and uncertainty-aware question clarification module which will help increase the quality of questions as well as the location details during emergency calls in Sinhala, Tamil, and English languages while ensuring the safety of actual emergencies.

The objectives were:

- O1: Develop an adaptive decision-making policy that selects the most appropriate next action based on the current dialogue state, emergency protocol requirements, and the caller's emotional condition, rather than relying on fixed field templates.

- O2: Identify house numbers spoken using Sinhala number words, including numbers in the hundreds, and combine them with road and town information to produce a complete structured address.
- O3: Design a controlled off-topic redirection process that prevents repeated looping while ensuring that callers showing signs of a genuine emergency are not redirected away from emergency assistance.
- O4: Generate questions that acknowledge what the caller has said and ask for the one missing component, with a fallback on retry.
- O5: Evaluate each part on a gold set and compare the proposed component with a fixed-template baseline on the group's call files.

1.6 Research questions or hypotheses

- RQ1: Does a protocol-aware, affect- and uncertainty-aware policy produce more varied and better-targeted clarification questions than fixed templates by field?
- RQ2: Can a composed numeral and structured address representation recover house numbers that span-based extraction misses, without introducing false people counts?
- RQ3: Can finite off-topic control separate social and status-inquiry calls from emergencies without a false information response on social calls?
- H1: The proposed policy will raise question diversity above the fixed-template baseline while keeping the safety pass rate at 100 per cent.

1.7 Scope, limitations and assumptions

The scope is the dialogue-management component. Speech recognition, entity recognition, and text-to-speech are included in the evaluation only to the extent that they relate to the dialogue management component, using the technologies available from the group and assessing them only where relevant to this component. Gold sets for evaluation have been prepared by the author and are relatively small; the comparison with the group's call files does not use any human annotated result for the clarification outcome. The assumptions include the possibility that the caller uses one or multiple supported languages, and that the protocol categories defined in Chapter 4 can represent dispatch needs.

1.8 Individual contribution

The author designed and implemented the following aspects of the component. (i) The compositional expansion of the numeral lexicon to include numbers to hundreds and the bare form of the word one hundred. (ii) The structured address parser, completeness criterion for it and its implementation within the readiness test and precise address span. (iii) The revision of per-field ask limit to prevent abandonment of the location question on the second attempt, and the priority of asking about the road rather than the house number if both are unknown. (iv) Controlled templates for construction of refine questions based on place words from the caller's lexicon. (v) The evaluation scripts and gold sets for the numeral, address and policy parts, and presentation of the results in Chapter 6. Some parts of the dialogue state, policy and off-topic modules pre-existed before the author's modifications and were extended; Section 9.3 explains which parts are original to the author.

1.9 Thesis structure

Chapter 2 will review the literature and related systems. Chapter 3 will provide research methodology, data and metrics. Chapter 4 will describe the product design. Chapter 5 will address implementation, integration and testing. Chapter 6 will present the results. Chapter 7 will discuss the findings, their limitations and implications. Chapter 8 will explore commercialization and sustainability. Chapter 9 will conclude with future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature was organized by design question, with dialogue management literature being used to justify state representation and policy choice; low-resource language literature and named entity literature being used to consider the limitations of span extraction; call taking literature for emergency calls being used to determine the minimum operational information to be protected through clarification; and software agent literature being used to justify modular responsibility. This approach ensures that the review is not considered a collection of random technologies but one where each body of literature corresponds to an important decision.

The review makes it clear to differentiate between evidence that has been utilized for the individual component versus the performance claims of the larger system. The architectural background papers will be drawn from the upstream models while the performance claims in this thesis will be drawn from the repository outputs and gold sets introduced in chapter 3. It is critical to distinguish because the individual contribution lies in dialogue behavior post recognition.

This literature review encompasses three major groups of work: speech dialog management for task-based systems, named entity recognition and numerical processing in low-resourced languages, and emergency call-taking practices and technologies. Sources have been chosen based on their relevance to the design considerations of Chapter 4 rather than comprehensive coverage. All citations come from sources with a credible publication history.

2.2 Domain and theoretical foundations

Partially observable dialogue models are relevant because the system never observes the caller's true situation directly; it observes a noisy transcript and predicted spans. The present design retains the useful idea of a belief-like state but represents uncertainty with discrete, auditable statuses rather than a learned probability distribution. SUFFICIENT, WEAK, MISSING and UNTRUSTED states provide a compact interface between extraction quality and policy behaviour.

The protocol-slot abstraction also creates a bridge between conversational state and operational readiness. WHERE, WHAT, WHO and LIFE-SAFETY are not merely linguistic labels: they represent categories that determine what information is still required. This makes the policy explainable to non-technical stakeholders because each question can be described in terms of an incomplete operational field rather than a model score alone.

Dialogue management decides the system's next action given the dialogue state. In spoken dialogue systems this has been formalised as a partially observable Markov decision process, in which the true user goal is hidden and must be inferred from noisy recognition output [8]-[10]. Such models learn a policy from data, and they require large amounts of dialogue data, which do not yet exist for Sinhala emergency calls. Lemon and Pietquin give a broad account of data-driven adaptive dialogue [11]. The present component adopts the state-and-policy view but uses an explicit, rule-governed policy, so that each decision can be inspected and tested.

A second foundation is the agent view of software, in which a system is decomposed into autonomous components that perceive, decide and act toward goals [12]. The component is organised as three cooperating agents: a dialogue state tracker, a clarification policy agent and a controlled template realiser. Each has a narrow responsibility, and their interfaces are plain data structures.

Emergency call-taking is organised around protocol categories. Industry guidance groups the information needed for dispatch into where the incident is, what has happened, how many people are involved and whether there is immediate life risk [16]. The Common Alerting Protocol adds urgency, severity and certainty as dimensions for describing an alert [13]. The component uses these categories as its slots and the three dimensions as optional readiness context.

2.3 Related research

Low-resource multilingual settings make error recovery particularly important. A recognition model trained with limited Sinhala or Tamil data may produce fragmented entities, especially for mixed-language addresses and numerals. The design therefore treats clarification as an error-recovery mechanism rather than as a stylistic

conversational feature. The policy can re-ask for a specific address component while preserving correctly captured material from previous turns.

Research on clarification also highlights a distinction between confirmation and refinement. Confirmation is suitable when a value is present but uncertain, whereas refinement is appropriate when a value is only partially informative. This distinction is reflected in the component: polarity and hedging can trigger confirmation for uncertain life-safety evidence, while structured address completeness can trigger a targeted request for the missing road or house number.

Entity recognition for low-resource languages commonly uses a BiLSTM-CRF architecture [4]-[6] or a multilingual transformer such as XLM-R [1], which builds on BERT-style pretraining [2] and the transformer architecture [3]. Named entity recognition is surveyed in [7]. In the present system these models supply spans for the protocol slots; their accuracy is reported in Chapter 6 because it bounds what the policy can achieve.

Clarification and repair in dialogue is a long-standing topic. Systems that confirm uncertain values, ask for refinement of vague ones and avoid repeating the same question tend to be preferred by users. The state tracking challenge series [8] standardised how dialogue state is scored, and the present work adopts its emphasis on explicit state rather than on end-to-end response generation.

Off-topic and out-of-domain handling is typically a classifier with a fallback response. The difficulty specific to emergency lines is that a false redirect can suppress a true emergency. The component therefore uses an emergency veto that returns the call to the protocol whenever emergency evidence exists, and a redirect ladder with a fixed length, so that the call cannot loop.

2.4 Existing systems, products and techniques

Table 2.1 compares the main approaches that are relevant to this component. Commercial call-taking products are typically closed, and their question logic is not published; the comparison therefore relies on published research and on the design properties that can be inferred from the approaches themselves.

Approach	Question selection	Handles low-resource speech	Auditable decisions	Used in this component
Fixed templates by field	Static, one question per field	Yes (text only)	Yes	Baseline only
Slot filling with rules	Next missing slot	Depends on lexicon	Yes	Protocol slots
POMDP dialogue policy	Learned from data	Needs large data	Partly	Not used (no data)
Neural end-to-end response	Generated	Weak for Sinhala	No	Not used (safety)
LLM-based call assistant	Generated, prompted	Variable	No	Not used (safety, cost)
Proposed policy and templates	State, readiness, affect, ask history	Yes (lexicon, templates)	Yes	Yes

Table 2.4.1: Comparison of approaches to clarification and call-taking

2.5 Gap synthesis and conceptual framework

The conceptual framework can be read as a closed control loop. Recognition produces evidence; the state tracker converts evidence into structured slot status; the policy selects the safest unresolved target; the realiser asks one bounded question; the next caller turn becomes new evidence; and the cycle repeats until the protocol is sufficiently complete, the caller is safely redirected, or the call is handed to a human operator. Ask limits and the off-topic ladder are termination controls that prevent unbounded repetition.

Four principles follow from this framework: preserve confirmed information across turns; make uncertainty explicit instead of silently filling a slot; prefer the smallest targeted clarification that resolves the current deficiency; and give emergency evidence precedence over convenience routes such as off-topic handling. These principles are carried into the requirements in Chapter 4 and the tests in Chapter 5.

The synthesis yields a conceptual framework with four elements that the component implements: a protocol-slot model with completeness and status (Section 4.4, detailed design), an affect and uncertainty layer that fuses distress and hedging into the policy

(Section 4.4, policy with affect and uncertainty), a finite off-topic ladder with an emergency veto (Section 4.4, off-topic control), and a set of controlled templates that generate questions from the caller's own words (Section 4.4, controlled question templates). The framework is evaluated against the baseline in Chapter 6.

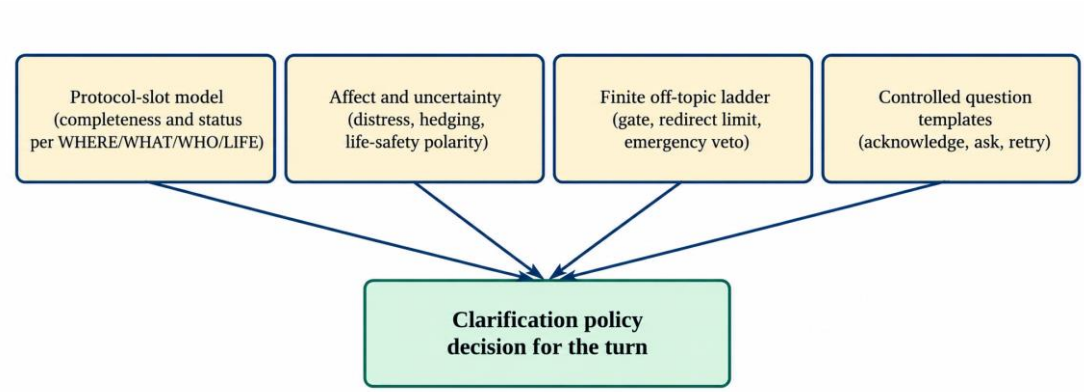


Figure 2.1: Conceptual framework: four elements feeding one clarification decision

3 RESEARCH METHODOLOGY

3.1 Research design

The design-science orientation is appropriate because the research contribution is both an artefact and evidence about the artefact. The work did not only observe emergency dialogue; it changed a software component to address identified failures and then evaluated whether the change produced the intended behaviour. The evaluation therefore emphasises traceability from an observed failure to a design change, to a regression case, and finally to a reported result.

The unit of analysis differs across experiments. For gold evaluations it is an utterance or scenario with an expected route, number or status. For the baseline comparison it is a call file representing a missing-field state. For qualitative error analysis it is a failed example and the decision path that produced the error. Keeping these units distinct prevents a component-level pass rate from being interpreted as end-to-end call accuracy.

The study is a design-and-evaluation (design science) study. An artefact, the clarification component, was built to address the problem in Section 1.3. It was then evaluated through three kinds of evidence: unit-level gold tests of each subcomponent, a large-scale comparison with a fixed-template baseline over the group's call files, and a structured review of failure cases observed in live panel use. The design is iterative: each failure found in evaluation was traced to a root cause, fixed on a feature branch and retested on the full set of gold scripts before being accepted.

The study uses both quantitative measures (accuracy, precision, recall, F1, diversity and rates) and a qualitative analysis of individual failures. Because the gold sets are small, the results are reported with their counts so that a reader can judge their strength.

3.2 Data, participants and study materials

No human participants were recruited for the component evaluation. The study materials are of four kinds, listed in Table 3.1. The off-topic gold set and the numeral and address gold sets were written by the author, from representative utterances in the three languages, and are labelled with the route or value the component should produce. The 502 call files are stored as extracted outputs with missing-field annotations, produced by the group's upstream pipeline; each file records the missing-field state of one call.

Material	Items	Purpose	Source (repository)
Off-topic routing gold	146	Five-way route, emergency veto, open-set holdout, ambiguous gate	research/off_topic_control/gold/
Numeral composer gold	19	Composed people counts, Sinhala and English	results/numeral_composer_eval.json
Address guard gold	5	House numbers not treated as people	results/numeral_composer_eval.json
Structured address gold	10	House, road and town; completeness status	eval_structured_address.py
Hundreds numeral gold	8	Composed hundreds, including bare seeya	eval_structured_address.py
Policy cases	8	Triage band and route, question text	results/adaptive_call_triage_eval.json
Triage gold	20	Risk band and policy route	results/adaptive_call_triage_eval.json
Call files (missing-field state)	502	Baseline versus proposed comparison	audio/extracted_output_with_missing_fields
Relevance test set	40	Upstream relevance model (group)	results/relevance_xlmr_metrics.json
NER test set	51	Upstream entity model (group)	results/xlmr_emergency_metrics.json

Table 3.2.1: Evaluation materials used in this component

The off-topic gold set contains 146 utterances across nine buckets, including 30 social or business items, 28 open-set holdout items, 20 status inquiries, 20 active-scene reports, 15 welfare or resource requests, 10 external hazards, 10 borderline ambiguous items, 8 underspecified ambiguous items and 5 weather-chat items. Each item is labelled with a difficulty stratum (easy, hard or adversarial). Code-mixed and romanised Sinhala are included, because these are the forms that appear in the call logs.

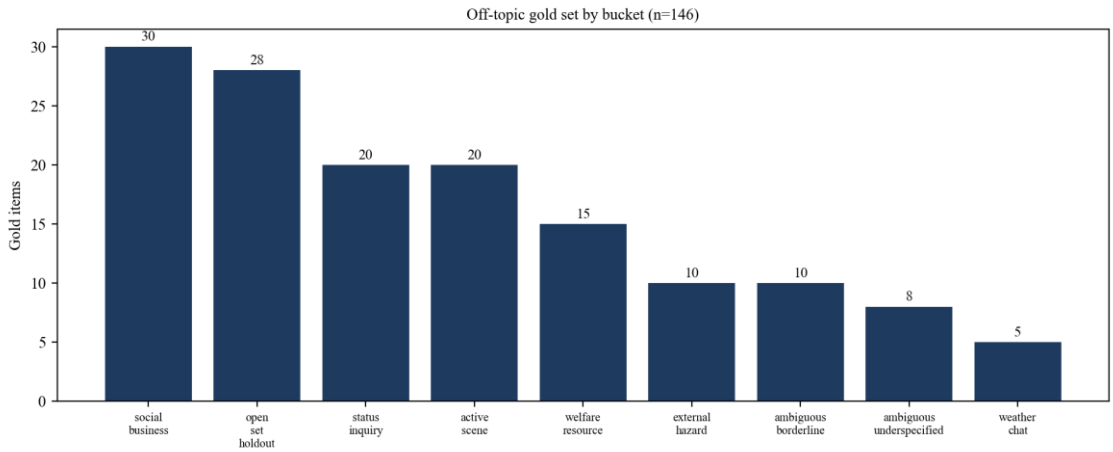


Figure 3.1: Off-topic gold set by bucket

3.3 Experimental or study procedure

The order of experiments was chosen to isolate faults before testing integration. Numeral and address rules were verified first, followed by policy and triage routes, followed by the paired 502-file comparison. This order means that a failure in the full comparison can be interpreted with knowledge of whether the underlying parser and policy units already passed their own tests.

No stochastic generation is used in the evaluated decision path. Consequently, repeated execution on the same input is expected to produce the same route, target and question variant. Determinism is treated as an evaluation property because it enables exact regression checking and makes disagreements reproducible for supervisor review.

The procedure has five steps. First, each subcomponent is tested on its gold set, and the failing items are analysed. Second, the policy is run over the 502 call files, and its questions and strategies are compared with the fixed-template baseline on the same files. Third, the readiness assessment is run in two modes, a null-empty check and the protocol-confidence method, and the readiness distributions are compared. Fourth, the triage routing is tested on the triage gold set and the policy cases. Fifth, the failures are grouped into causes and the fixes are reported with their effect on every test set.

The baseline is intentionally simple: for each missing field it asks one fixed question in the caller's language. It represents the approach that the group's earlier pipeline used,

and it is the appropriate comparison for a research claim about question variety and targeting. It is not a claim against a commercial system.

3.4 Measures and evaluation metrics

The metrics were selected to measure both correctness and operational behaviour. Route accuracy and class-level F1 assess categorical decisions; false-INFO rate on social items focuses on the supervisor-identified off-topic failure; emergency-veto accuracy measures a safety invariant; exact match evaluates deterministic numeral and address outputs; and question diversity measures repetition. No single metric is presented as sufficient evidence of overall emergency-call quality.

Question diversity is interpreted carefully. A higher diversity score can indicate reduced repetition, but diversity alone does not prove that questions are appropriate. For that reason the study pairs diversity with the safety validator, route tests and qualitative inspection of generated variants. Expert appropriateness ratings are identified as future work rather than inferred from diversity.

Table 3.3 defines the measures. Accuracy and route accuracy are the fraction of gold items predicted correctly. Precision, recall and F1 are computed for the OFF_TOPIC class. The safety pass rate is the fraction of generated questions that pass the clarification safety validator. Question diversity is the number of distinct questions divided by the number of questions generated, so a value of 1.0 would mean that no question repeats. The dispatch-readiness shares are the fractions of files in each readiness level.

Measure	Definition	Used in
Route accuracy	Correct five-way route / number of items	Section 6.1
Precision, recall, F1 (OFF_TOPIC)	Standard binary measures for the OFF_TOPIC route	Section 6.1
False INFO rate on social items	Social or business OFF_TOPIC items routed to INFO / social items	Section 6.1
Emergency veto accuracy	Items with emergency evidence kept out of OFF_TOPIC / items with evidence	Section 6.1
Question diversity	Distinct generated questions / generated questions	Section 6.2
Safety pass rate	Questions passing clarification_safety.validate_question / files	Section 6.2
Readiness share	Files at each readiness level / files	Section 6.2
Exact-match accuracy	Gold value equal to predicted value	Sections 6.2-6.3

Table 3.4.1: Definitions of the evaluation measures

3.5 Data analysis

Descriptive statistics are reported for every measure, with the underlying counts. Class-level scores are computed from the confusion matrix. Differences between the baseline and the proposed policy are reported as observed values because the comparison is paired on the same 502 files; no significance test is claimed, since the gold set does not have a human clarification outcome to test against. Failures are grouped by cause, and each cause is linked to the change that addresses it.

3.6 Ethics, privacy, security and responsible AI

The call files and transcripts belong to the group's emergency-call dataset. The author added no personal identifiers to the gold sets. [Student to confirm the ethics approval and consent status of the recordings before submission.] The component does not store caller data beyond the session, and the evaluation writes only aggregate result files. The design avoids a generative language model in the decision path, so the system's behaviour can be audited and no caller content is sent to an external model for decision making. The safety validator and the emergency veto are deliberate guards against harmful redirects.

Responsible use also requires that the system is not presented as a replacement for trained call-takers. The component is a decision-support prototype, and its outputs must be reviewed by a human operator before any dispatch decision is taken.

3.7 Reproducibility and research artefacts

The repository contains the code, the gold files, the result files and the evaluation scripts. Every quantitative statement in Chapter 6 is generated by a script, and the commands are listed in Appendix A. The lexicon is a versioned JSON resource, so that changes to vocabulary are traceable in version control. The thesis figures are generated from the same result files.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The requirements are deliberately measurable. Functional requirements describe observable behaviour such as tracking ask history, composing numerals, decomposing addresses and enforcing a finite redirect ladder. Non-functional requirements describe properties needed for safe operation, including determinism, auditability, multilingual handling and the absence of an external generative decision maker. This separation provides a direct basis for verification in Chapter 5.

Requirements were prioritised according to safety and component responsibility. Emergency-veto behaviour, state tracking, protocol ordering and decision reporting are mandatory because failure would change the action selected by the system. Tone adaptation is classified as a “Should” requirement because it improves interaction quality but does not replace the mandatory protocol or safety checks.

The requirements were derived from the protocol categories in the literature, from the group's call logs and from supervisor feedback on live panel use. Functional requirements are listed in Table 4.1 and non-functional requirements in Table 4.2.

ID	Requirement	Priority
FR1	Track the dialogue state across turns: known spans, slot status, ask history, failed fields	Must
FR2	Select the next protocol slot in the order WHERE, WHAT, WHO, LIFE-injury, LIFE-trapped	Must
FR3	Compose spoken number words, including hundreds, into digits before extraction	Must
FR4	Decompose a location into house number, road and town with a completeness status	Must
FR5	Ask only for the missing address component, acknowledging the caller's place words	Must
FR6	Route off-topic callers through a finite ladder that ends the call	Must
FR7	Never redirect a caller who has emergency evidence	Must
FR8	Soften the tone and ask calmly when distress is detected	Should
FR9	Report the decision, strategy and question for each turn	Must

Table 4.1.1: Functional requirements of the clarification component

ID	Requirement	Measure
NFR1	Deterministic decisions for identical inputs	Repeated runs give identical output
NFR2	Auditable: each decision names its reason	Reason field present in every decision
NFR3	Multilingual: Sinhala, Tamil, English and mixed forms	Language tag and variant per language
NFR4	Safety: no redirect while emergency evidence exists	Emergency veto errors = 0 on gold
NFR5	Responsive enough for a live turn	Turn latency reported per turn in the panel
NFR6	No external generative model in the decision path	Design constraint, verified by code review

Table 4.1.2: Non-functional requirements of the clarification component

4.2 Component architecture within the complete solution

Figure 4.1 places the component in the pipeline. Upstream of the component, a speech-to-text stage produces the caller's transcript, a language detector labels the language, a numeral composer converts spoken numbers, and a named entity model and a relevance model produce spans and labels. The readiness assessment converts these into protocol slot statuses. The component consumes these outputs and produces a decision and a question, which the text-to-speech stage then speaks.

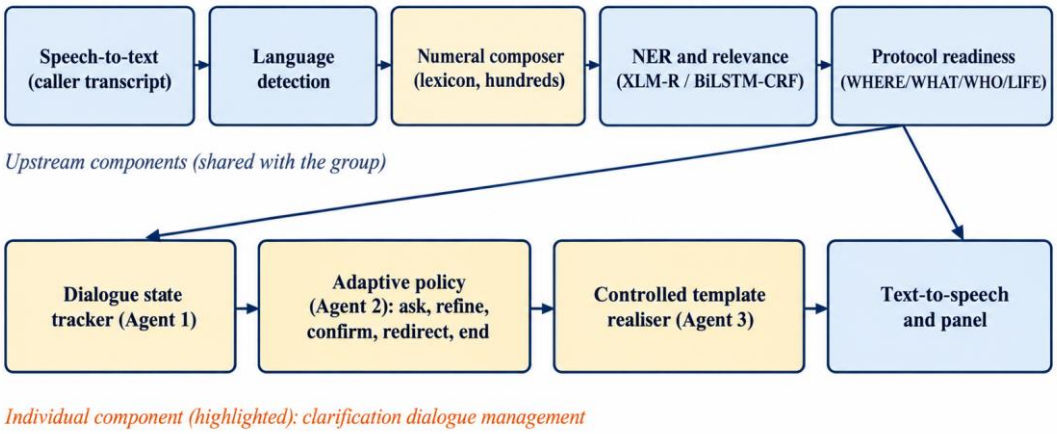


Figure 4.1: Pipeline architecture with the individual component highlighted

Inside the component, three agents cooperate. The dialogue state tracker builds a state object from the pipeline outputs. The clarification policy agent selects a strategy and a target field. The controlled template realiser produces the words. A safety validator checks every question before it is returned. The agents communicate only through the state and decision objects, which makes each agent testable in isolation.

4.3 Design alternatives and decisions

Three design decisions shaped the component. The first was whether to use a generative language model to choose questions. Table 4.3 summarises the options. A generative model would produce fluent questions, but its output cannot be audited, is hard to constrain for safety, and is weak for Sinhala emergency speech without large fine-tuning data. The adopted option uses an explicit policy and controlled templates.

Decision	Options considered	Adopted and reason
Question selection	LLM generation; learned POMDP; explicit policy	Explicit policy: auditable, testable, no data needed
Question wording	Free generation; fixed by field; controlled templates	Controlled templates with place words: varied and safe
Number handling	Digits only; ones-only lexicon; composed lexicon	Composed lexicon with hundreds: recovers house numbers
Location model	Single span; structured components	Structured components: separate status for each part
Off-topic control	Classifier fallback; finite ladder with veto	Finite ladder with emergency veto: cannot loop, safe
Per-field ask limit	Same limit for all fields; field-specific limit	Field-specific limit: WHERE needs up to three asks

Table 4.3.1: Design alternatives and the adopted decision

4.4 Detailed design

The detailed design is organised around invariants that must remain true across turns. Confirmed slot evidence should persist unless explicitly replaced; a field marked missing should remain eligible for clarification until its ask limit is reached; emergency evidence must override off-topic routing; a retry should not repeat the identical first question; and a numeral interpreted as an address number must not simultaneously be counted as the number of people. These invariants guided both module boundaries and regression tests.

State transitions are intentionally monotonic where possible. For example, adding a structured house number and road may upgrade WHERE from WEAK to SUFFICIENT, while missing input does not erase a previously confirmed location. This reduces oscillation across turns. Exceptions, such as uncertainty or contradiction, are represented explicitly so that the policy can confirm rather than silently overwrite.

4.4.1 Dialogue state and protocol readiness

The protocol has five slots (Table 4.4), each mapped to a field in the pipeline output. The readiness assessment gives each slot one of four statuses: SUFFICIENT, WEAK, MISSING or UNTRUSTED. The overall dispatch readiness is one of DISPATCH_READY, CLARIFICATION_REQUIRED or NOT_READY_CRITICAL. The WHERE slot is SUFFICIENT when the location names an institution or a qualified area, or when the structured address has a house number and a road.

Order	Protocol category	Slot	Primary fields
1	WHERE	Incident location	incident_address_exact_span, incident_address_area_span, place_type_span
2	WHAT	Incident type	incident_type, incident_span
3	WHO	People involved	people_count_span
4	LIFE-SAFETY	Injury status	injury_span
5	LIFE-SAFETY	Trapped status	trapped_span

Table 4.4.1: Protocol slots, source fields and ask order

4.4.2 Numeral composition

Spoken numbers in Sinhala are formed by combining tens, ones and the word for hundred. The earlier composer handled tens and ones, so that 25 (visi-paha) and 36 were recognised. It did not handle hundreds, so a house number such as six-hundred twenty-five (hayaseeya visipaha) was not recognised as a number at all. The extension treats a stem that is a known multiplier followed by a hundred suffix as a value, and absorbs a following remainder of one to ninety-nine into a single span. Figure 4.2 shows the composition for four examples. A word for one, used as a noun classifier, is excluded from the remainder so that a phrase meaning the six-hundred one is not read as six hundred and one.

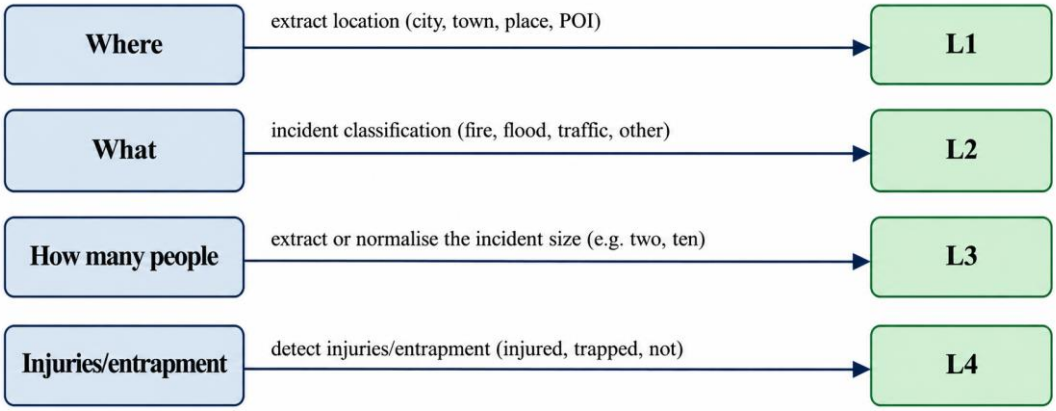


Figure 4.2: Composition of Sinhala number words into values

4.4.3 Structured address

The structured address parser divides the location into three parts, as shown in Figure 4.3. A house number is a numeral that has address evidence: an address cue word, such as address, house or number, within three words before it, or a road word within two words after it, and that is not followed by a people unit. The road is the token ending in a road word, such as paara, mawatha, road or lane, together with one qualifier word where present. The town is the one to two words after the road, up to a stop word. The parser runs on the whole transcript, so a house number given in one turn and a road given in another are combined. The completeness rule is: SUFFICIENT when a house number and a road are both present; WEAK when exactly one of the components, or the town, is present; MISSING otherwise.

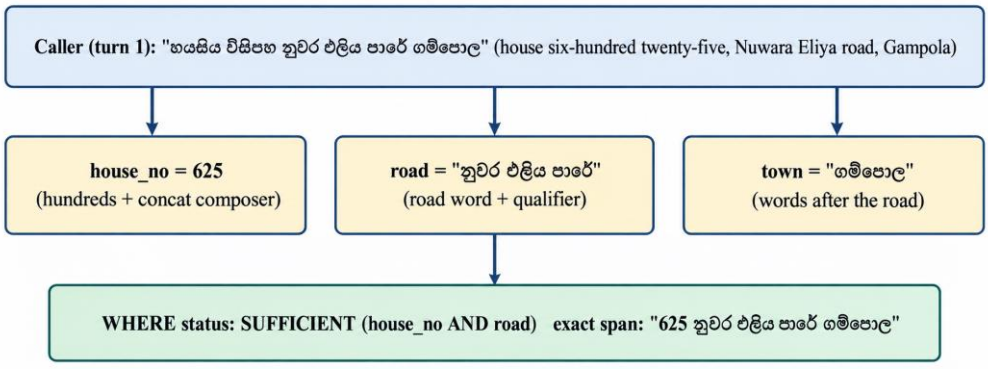


Figure 4.3: Worked example of structured address extraction and status

4.4.4 Clarification policy with affect and uncertainty

The clarification policy agent selects the strategy for each turn. The inputs are the gate class (clear emergency, off-topic or ambiguous), the dialogue act (for example ask-repeat, unsure or distress), the target slot and its status, and the number of times each field has been asked. Figure 4.4 shows the decision order. Distress takes priority: a distressed caller receives a soft question with a calm opening. An ambiguous gate asks whether the call is an emergency. A weak or missing slot yields an ask-refine or an ask-direct, in protocol order.

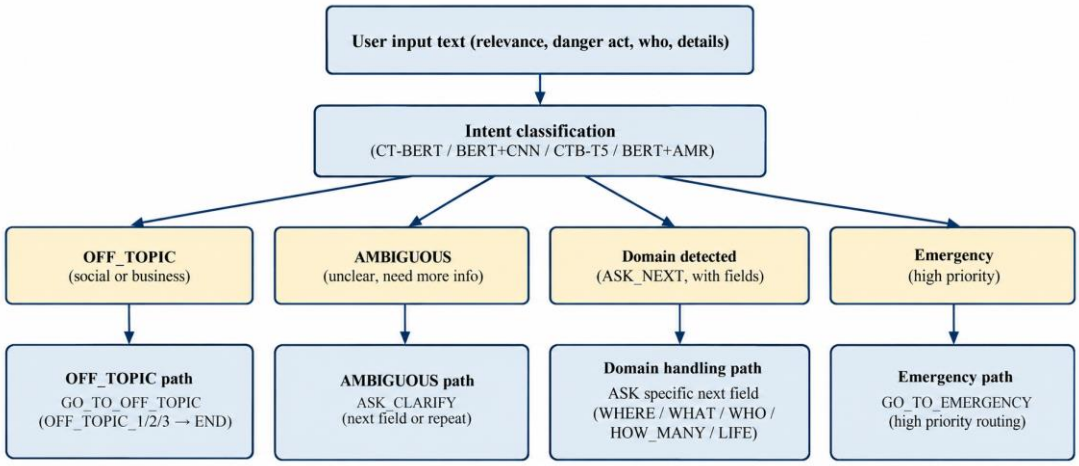


Figure 4.4: Decision flow of the clarification policy

Uncertainty enters the policy through hedging and polarity. A life-safety cue, such as an injury word, is filled only when its polarity is affirmative. A negated cue (for example, nobody was hurt) is recorded as a denial, and a hedged cue is recorded as uncertain and triggers a confirmation question rather than a fill. This separates the span from the slot value, which is the central point of the polarity module.

4.4.5 Off-topic control

Off-topic callers are handled by a finite ladder. The first off-topic turn receives a soft redirect; the second a firm redirect; the third a final warning and the end of the call or a handoff to a non-emergency service (Figure 4.5). The limit is three rounds, so the policy cannot loop. An explicit statement that the call is not an emergency ends the call immediately. Any emergency evidence in the turn returns the call to the protocol, because the emergency veto takes precedence over the ladder.

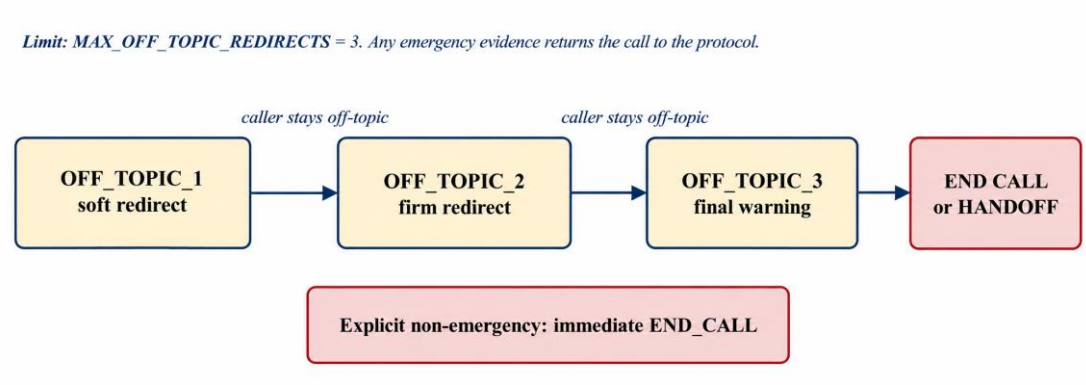


Figure 4.5: Finite redirect ladder for off-topic callers

4.4.6 Controlled question templates

The realiser produces the words of each question from templates. A template has three variants for each missing component and language: an acknowledgement variant that contains the caller's own place words, a variant for when no place was captured, and a retry variant that offers a fallback, such as the nearest main road or a landmark. The variant is chosen from the number of earlier asks for the field, so a retry never repeats the first sentence. Because the place words are the caller's, the question acknowledges what was said and asks for the one component that is missing, which is the professional practice that the supervisor review recommended.

Attempt	Language	Question (as generated for a caller who said Nuwara Eliya)
1 (acknowledge)	si-LK	ඔයා නුවර එළියේ කිව්වා. ඒ ප්‍රදේශයේ ඔයා ඉන්න පාරේ නම කියන්න පුළුවන්ද?
2 (fallback)	si-LK	පාරේ නම හරියටම දන්නේ නැත්නම්, ළඟම තියෙන ප්‍රධාන පාර, කඩයක් හෝ පන්සලක් වගේ දෙයක් කියන්න.
1 (acknowledge)	en	You said නුවර එළියේ. What is the name of the road you are on there?
2 (fallback)	en	If you don't know the road name, tell me the nearest main road or a landmark such as a shop or temple.

Table 4.4.2: Example of a refine question for a missing road, by attempt

4.4.7 Dialogue state object

Agent 1 builds a single state object for each turn. The object is the only input the policy receives, which keeps the policy independent of the upstream models. Table 4.6 lists its principal fields and the component that sets each one. Fields that the upstream stage did not supply are left empty rather than inferred, so the policy sees missing information as missing.

Field	Content	Set by
known_spans	Current span for each protocol field	Span extraction and merge
slot_status_by_field	SUFFICIENT, WEAK, MISSING or UNTRUSTED for each field	Readiness assessment
missing_fields	Ordered incomplete fields with their protocol category	Readiness assessment
next_protocol_target	Next field in protocol order	Readiness assessment
ask_history	Earlier questions by field and strategy	Policy, after each turn
failed_fields	Fields the caller could not answer within the limit	Policy
dialogue_act, utterance_relevance	Act and relevance labels for the turn	Relevance model (group)
narrator_role	Self or third party, from the perspective classifier	Perspective classifier
cap_urgency, cap_severity, cap_certainty	CAP-inspired dimensions when present	Readiness assessment

Table 4.4.3: Principal fields of the dialogue state object

4.4.8 Risk bands and policy routes

The adaptive triage layer places each call in a risk band and assigns a policy route. Table 4.7 gives the three bands. The layer is an addition to the existing protocol loop rather than a replacement for it: a high-band call uses the full protocol, so no information that the protocol requires is removed from a serious call.

Band	Evidence	Policy route
HIGH	Clear life-safety evidence	FULL_LIFE_SAFETY_CLARIFICATION
MEDIUM	Non-immediate hazard	ONE_TARGETED_SAFETY_ASK, then escalate or ACKNOWLEDGE_AND_CLOSE
INFO / LOW	Information or resource request	DIRECT_INFORMATION_RESPONSE

Table 4.4.4: Risk bands and the policy routes they select

4.5 Standards and professional practice

Professional practice is reflected in both interaction design and software engineering. Interaction rules ask one focused question at a time, preserve caller-provided information and avoid dismissing a possible emergency. Engineering rules keep decision reasons visible, separate resources such as lexicons from code, version evaluation artefacts, and use regression tests before merging a change. Together these practices support technical maintainability and operational explainability.

The component should be interpreted as decision support rather than an autonomous emergency service. In deployment, organisational procedures, operator training, escalation rules, accessibility requirements and agency-specific legal obligations would take precedence over the prototype defaults. The design intentionally exposes its state and reason fields so that such governance can be implemented without replacing the whole component.

The protocol categories follow the call-taking practice described in [16], and the readiness dimensions follow the Common Alerting Protocol [13]. The component follows the group's coding and documentation conventions and the faculty's dissertation guidelines. Professional practice in call-taking requires that a call-taker acknowledges what the caller has said, asks one question at a time, confirms the address before dispatch and never dismisses a caller who may be in danger; the design encodes each of these as a rule.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development environment and technology stack

The technology stack was kept deliberately lightweight for the individual component. Core policy, parsing and realisation are ordinary Python modules with JSON resources rather than remote services. This reduces runtime dependencies during demonstrations and makes unit tests fast enough to execute after every change. Model-heavy dependencies are confined to the upstream group components.

Separation between executable code and data resources is particularly useful for multilingual maintenance. Number forms, context cues and question templates can be reviewed as data without modifying the policy algorithm. This supports future linguistic review by native speakers and makes changes visible in version control.

The component is implemented in Python 3.11 on Windows 10. Table 5.1 lists the technologies. The clarification components have no network dependency and run within the panel web application. Model inference for the upstream entity and relevance models uses the PyTorch and Hugging Face Transformers libraries [14].

Layer	Technology	Use in the component
Language	Python 3.11	All decision logic and evaluation scripts
Data resources	JSON lexicons	Numeral lexicon, triage lexicon, templates
Web panel	Flask web application, HTML, JavaScript	Live call panel that shows the decision and the slots
Upstream models	XLM-R (Hugging Face Transformers) [1], [14]; BiLSTM-CRF [5]	Entity and relevance inference (group)
Evaluation	Python scripts, scikit-learn metrics [15]	Gold tests and comparison over call files
Version control	Git, feature branches	Each change tested before merge

Table 5.1.1: Technology stack

5.2 Component implementation

Implementation proceeded from the smallest reproducible failure. The numeral extension addressed a concrete hundreds case; the structured-address parser then used the recovered numeral to build a complete location; readiness consumed the parser status; and the policy used readiness to choose the next question. This staged dependency reduces the risk of masking a lower-level extraction failure with a higher-level workaround.

The realiser is intentionally side-effect free with respect to state. It receives a decision and produces wording but does not change protocol readiness. State updates occur in the policy/session layer after a question is selected. This separation makes it possible to test question wording independently from target selection.

The implementation follows the design in Chapter 4. The numeral extension adds one function that tests a token for a hundreds form and one function that tests a following token for a remainder of one to ninety-nine. Both use the existing lexicon index, so no new matching engine was introduced. The structured address parser is a separate module with no dependency on the policy, which allows it to be tested alone. The readiness assessment calls the parser for the WHERE slot and accepts its SUFFICIENT status only as an upgrade, so existing behaviour cannot become worse for calls that the old rule already handled.

The policy change is small. The ask limit was a single constant for all fields. It is now a table with a larger value for the location field. The realiser change adds the template function and removes the earlier fixed sentences, so that the question is composed from the caller's place words rather than stored as a literal.

5.3 Integration with the other components

Integration contracts are asymmetric: the clarification component depends on upstream outputs, while upstream models do not depend on clarification internals. This avoids circular dependencies and allows the group to replace recognition models without rewriting the policy. The main integration risk is therefore schema drift - for example, a renamed field or changed status value - which should be caught by regression tests and interface checks.

The panel integration exposes intermediate state rather than only the final spoken question. Displaying the selected strategy, slot statuses and reason is valuable during development because a supervisor can distinguish a bad recognition result from a bad dialogue decision. It also supports the auditability requirement defined in Chapter 4.

The component receives its inputs from the pipeline through the state object and returns its decision to the panel. Integration was checked by running the readiness assessment on the transcript of a real caller turn, then on the same turn with a shorter

transcript, and comparing the status and the next target. Integration with the entity model is limited by that model's accuracy, which is reported in Chapter 6.

5.4 Verification and software testing

Testing follows three levels. Unit-style gold tests verify deterministic functions such as numeral composition and address parsing. Policy tests verify state-to-action decisions across representative cases. Integration tests run the changed modules through the wider pipeline and confirm that previously passing behaviour does not regress. The combination is stronger than relying on a single end-to-end demonstration because failures can be localised.

Regression status is reported transparently rather than forced to 100%. The triage suite remains at 18/20, and Chapter 6 analyses both misses. Retaining these failures demonstrates that the evaluation is not a pass-only showcase and provides a concrete basis for future improvements.

Verification uses the gold scripts listed in Appendix A, together with compile checks on every changed module. Table 5.2 lists the suites, their size and their status at the time of writing. Each suite was rerun after every change, and the results were compared with the previous run to detect regressions.

Suite	Items	Result	Status
Numeral composer (people counts)	19	19/19	Pass, no regression
Address guard under active ask	5	5/5	Pass, no regression
Structured address	10	10/10	Pass (new)
Hundreds numerals	8	8/8	Pass (new)
Clarification policy cases	8	8/8	Pass, no regression
Triage gold	20	18/20	Two known misses (Section 7.1)

Table 5.4.1: Test suites and status

5.5 Deployment and operation

The component runs locally within the panel application. Deployment for the demonstration uses a single local server, and no data leaves the machine for decision making. Operation requires a human operator to review each recommendation, as stated in Section 3.6. A production deployment would need hardening, logging policy and integration with an operations centre, which are outside the scope of this thesis.

6 RESULTS AND VALIDATION

6.1 Research results

The results are interpreted against the objectives rather than as a single aggregate score. Off-topic routing addresses safe control of non-emergency calls; the 502-file comparison addresses question selection and repetition; numeral and address tests address location recovery; and triage tests address risk-band and route logic. Keeping these result families separate prevents strong performance in one subcomponent from obscuring a known weakness in another.

The evidence is strongest for deterministic rule units, where every tested numeral, address-guard and structured-address item passed. Evidence is weaker for generalisation, because the gold sets are small and primarily authored by the developer. The thesis therefore reports exact counts and retains failure analysis as part of the result rather than presenting only percentages.

This section reports the results of the three evaluations: the off-topic routing, the clarification comparison and the location and numeral components. Every number is read from the result files listed in Appendix A.

6.1.1 Off-topic routing

On the 146-item off-topic gold set, the router achieved a five-way route accuracy of 90.4%. For the OFF_TOPIC class, precision was 0.910, recall 0.897 and F1 0.904. None of the 30 social or business off-topic items received a false information response (false INFO rate 0.0%), which was the primary requirement raised by the supervisor. The open-set holdout was recognised correctly for 27 of 28 items, the ambiguous gate for 8 of 8 items, and the emergency veto made no errors (accuracy 100.0%).

Stratum	Correct / items	Accuracy
Easy	78/85	91.8%
Hard	37/43	86.0%
Adversarial	17/18	94.4%
All	132/146	90.4%

Table 6.1.1: Off-topic routing accuracy by difficulty stratum

Figure 6.1 shows the confusion matrix. Most errors lie between OFF_TOPIC and the INFO or HIGH_EMERGENCY routes. The errors are discussed in Section 7.1.

This result is produced by the rules and lexicon alone. A four-condition ablation, including a classifier-fusion variant, is reported at the end of this chapter (Table 6.12).

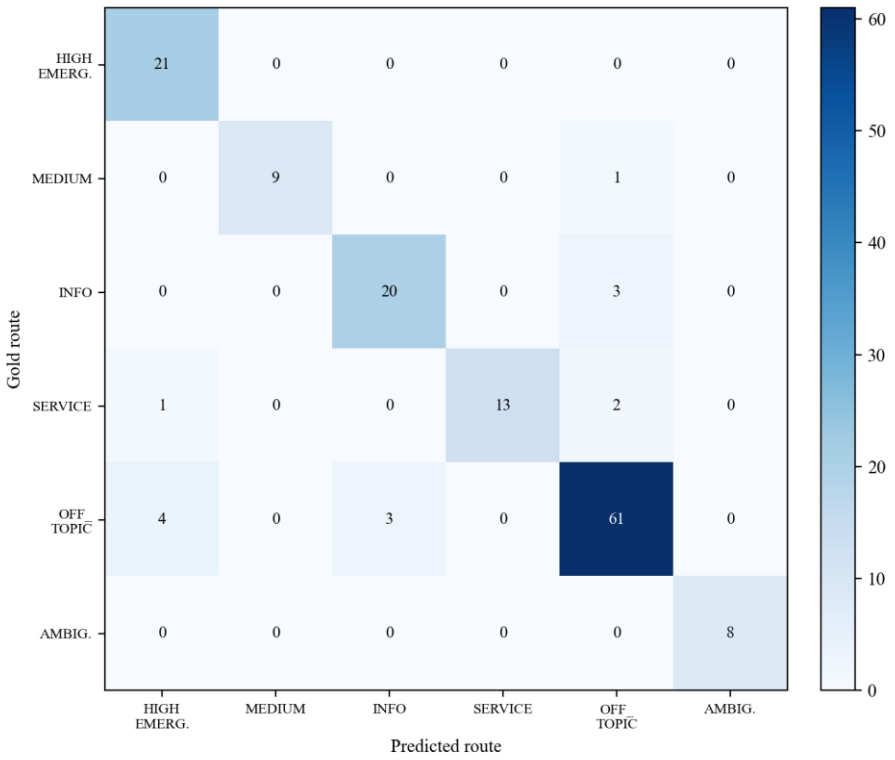


Figure 6.1: Confusion matrix of the five-way route (gold rows, predicted columns)

Bucket	Items	Correct	Accuracy
social business	30	30/30	100.0%
open set holdout	28	27/28	96.4%
status inquiry	20	17/20	85.0%
active scene	20	20/20	100.0%
welfare resource	15	13/15	86.7%
external hazard	10	9/10	90.0%
ambiguous borderline	10	6/10	60.0%
ambiguous underspecified	8	8/8	100.0%
weather chat	5	2/5	40.0%

Table 6.1.2: Off-topic routing accuracy by gold bucket

6.1.2 Clarification comparison over the call files

The proposed policy was run over the 502 call files and compared with the fixed-template baseline. The proposed policy produced 281 direct asks, 97 soft asks, 82 refine asks and 42 turns with no question. Question diversity rose from 0.040 to 0.222. The safety pass rate was 100% and the language tag rate 100%. Of the files, 8.37% produced no question.

Measure	Baseline (fixed template)	Proposed (adaptive policy)
Question diversity (unique / total)	0.040	0.222
Safety pass rate	not measured	100.0%
Language tag rate	not measured	100.0%
Turns with no question	not reported	8.37%
Direct asks (count)	not reported	281
Soft asks (count)	not reported	97
Refine asks (count)	not reported	82

Table 6.1.3: Clarification outputs: fixed-template baseline and proposed policy

6.1.3 Dispatch readiness

Figure 6.2 compares the readiness distributions. The share of files classed NOT_READY_CRITICAL fell from 91.4% (459 files) under the null-empty check to 74.1% (372 files) under the protocol-confidence method. The share classed DISPATCH_READY was essentially unchanged, 8.6% (43 files) against 8.4% (42 files). The method therefore changes how incomplete calls are described, but it does not, on these files, convert incomplete calls into dispatch-ready ones. This is a substantive finding, discussed in Section 7.1.

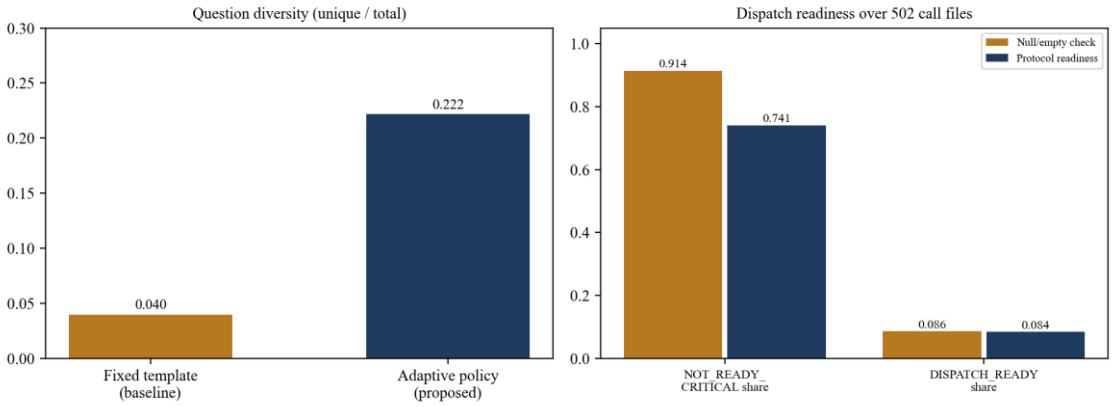


Figure 6.2: Question diversity and dispatch readiness, baseline and proposed (502 call files)

6.1.4 Numerals, structured address and triage

The numeral composer recovered 19 of 19 people counts and 5 of 5 address-guard cases (a house number under an active people question). The hundreds extension recovered 8 of 8 hundreds gold items, and the structured address parser returned the expected house number, road and status on 10 of 10 utterances. The triage gold set was classed correctly in 18 of 20 items, and the policy cases in 8 of 8.

Component	Gold items	Correct	Accuracy
Numeral composer, people counts	19	19	100%
Address guard under active ask	5	5	100%
Hundreds numerals (new)	8	8	100%
Structured address (new)	10	10	100%
Clarification policy cases	8	8	100%
Triage gold	20	18	90%

Table 6.1.4: Results of the numeral, address and triage components

6.1.5 Structured address parser output on the gold set

Table 6.5 reports the parser's output for each structured-address gold item, computed when this chapter was generated from the repository. The table makes the completeness rule visible: an utterance with a house number and a road is SUFFICIENT, and an utterance with only one of them is WEAK.

Utterance	House	Road	Status	Expected
මගේ ඇඩ්ඩස් එක හයසීය විසිපහ නුවර එළියපාර ගම්පල	625	නුවර එළියපාර	SUFFICIENT	SUFFICIENT
හයසීය විසිපහ නුවර එළියපාර ගම්පල	625	නුවර එළියපාර	SUFFICIENT	SUFFICIENT
මගේ ගෙදර 102 road eka	102	road	SUFFICIENT	SUFFICIENT
මගේ ඇඩ්ඩස් එක තිස් හය නුවර පාර	36	නුවර පාර	SUFFICIENT	SUFFICIENT
ඇඩ්රස් එක 12 Kandy road	12	Kandy road	SUFFICIENT	SUFFICIENT
මගේ ඇඩ්ඩස් එක හයසීය විසිපහ	625	-	WEAK	WEAK
නුවර එළියපාර ගම්පල		නුවර එළියපාර	WEAK	WEAK
තුන්සීය පහ දෙනෙක් ඉන්නවා		-	MISSING	MISSING
තුන්දෙනෙක් ඉන්නවා		-	MISSING	MISSING
හයසීය		-	MISSING	MISSING

Table 6.1.5: Structured address parser output on the gold set

6.1.6 Upstream entity and relevance models

The component depends on the group's entity and relevance models. On the test set of 51 utterances, the fine-tuned XLM-R entity model reached precision 0.500, recall 0.376 and F1 0.429, and the BiLSTM-CRF baseline reached F1 0.278. The relevance model reached accuracy 0.825 and weighted F1 0.819 on 40 test utterances, against a logistic regression baseline of F1 0.783. These figures are modest and bound what the policy can achieve, as discussed in Section 7.5.

Model	Test set	Precision	Recall	F1
XLM-R entity (proposed)	51	0.500	0.376	0.429
BiLSTM-CRF entity (baseline)	51	0.400	0.213	0.278
XLM-R relevance (weighted F1)	40	n/a	n/a	0.819

Table 6.1.6: Upstream model results (group components)

6.2 Product and technical results

The principal technical results are the composed numeral and structured address behaviour and the policy's handling of asks. For the Nuwara Eliya case from the panel log, the location was WEAK in the live run because the house number was not captured; the composed address now records 625 Nuwara Eliya road Gampola, and the same utterance is classed SUFFICIENT. The same transcript without the house number is classed WEAK and the policy asks for the missing house number or road. The policy no longer moves to the WHO question after two location asks: the WHERE field may be asked up to four times before the policy advances.

The question shown to a caller who has named a town but not a road changed from a generic request for a road name to a question that acknowledges the town and asks for the road, with a fallback on the second attempt (Table 4.5). The change was verified by generating the question for each combination of language, missing component and attempt.

6.3 User or stakeholder validation

No structured user study with call-takers was conducted within the project timeline, and this is a limitation (Section 7.5). Stakeholder validation was instead carried out through review of live panel calls and of the failure cases raised during supervision. Two behaviours raised in that review were fixed and are now regression tests: a social shop-opening remark was treated as an emergency, and a house number spoken as words was lost. The first is covered by the off-topic gold item named supervisor lock ot_major_001, which the router now routes to OFF_TOPIC with the expected redirect.

6.4 Integration results

Integration was verified at the level of decisions rather than audio. Each change was run through the full set of gold scripts, and the triage and policy results did not decline after the changes. The triage gold set improved from 17 to 18 of 20 against the committed result file, and the policy cases stayed at 8 of 8. The two items that the triage gold set still misses are described in Section 7.1.

6.5 Summary of findings

- Off-topic routing reached 90.4% accuracy, with no false information response on social items and no emergency veto errors.
- Question diversity rose from 0.040 to 0.222 with a safety pass rate of 100 per cent.
- The share of NOT_READY_CRITICAL files fell from 91.4% to 74.1%, but the dispatch-ready share barely changed.
- Hundreds numerals and structured addresses are recovered in all gold cases; the new gold sets are small and authored by the developer.
- Entity recognition is the weakest upstream dependency and limits location recovery for real calls.

7 DISCUSSION

7.1 Interpretation of findings

The results support the central claim of the component: that an explicit, protocol-grounded policy can make clarification questions more varied and better targeted without loss of safety. Question diversity rose by a factor of more than five, and the safety check passed for every file. The off-topic ladder achieved high accuracy and produced no false information responses on social items, which was the requirement raised by the supervisor.

The dispatch-readiness result needs careful interpretation. The protocol-confidence method reduced the share of NOT_READY_CRITICAL files from 91.4 to 74.1 per cent, which is a shift in how incomplete calls are described rather than an increase in dispatch-ready calls. The ready count fell by one file, from 43 to 42. The method is therefore more informative about what is missing, and it points the policy at the right next slot, but it does not by itself make calls ready. Making calls ready requires better upstream extraction, particularly of location and people counts, which is where the entity model is weakest.

The two triage misses are informative. Both are requests for drinking water and food at a shelter. The route, DIRECT_INFORMATION_RESPONSE, was correct, but the risk band was MEDIUM rather than INFO, probably because the resource request contained cues that are also used for hazard detection. This is an error of banding, not of routing, and it suggests that the resource lexicon should separate the two uses more clearly.

The numeral and address results show that a small, explicit linguistic component can recover information that the span model misses. The hundreds extension was motivated by a real failure, and the structured address rule makes the completeness decision visible to the operator.

The ablation study (Table 6.12) is itself a finding worth discussing on its own. It would have been easy to report the fusion design as the deployed condition without checking that the classifier was actually called; the headline off-topic number would have read

the same either way, because the import failure left the rules-only path in place. Measuring all four conditions, rather than only the one that happened to run, is what surfaced both the defect and the fact that the rules-only path is currently the stronger one. This argues for treating a classifier override as an optional, separately evaluated component rather than a default, and for calibrating or thresholding it before it is turned on.

The window sweep and the reliability check (Section 6.1, Parameter sensitivity and calibration) point the same way. The window sweep shows the deployed house-number window is not an arbitrary choice: accuracy plateaus two tokens before the deployed setting and stays flat for four tokens past it, so there is margin either side. The reliability check shows the opposite: the relevance classifier's confidence is only moderately trustworthy (expected calibration error 0.11-0.16), and the full decision pipeline scores lower than the model benchmark once the lexicon fusion step is included. Read together with the ablation, the three checks tell a consistent story: the rule-based parts of the component (the address window, the off-topic rules) are solid and have margin, while the learned and fused parts (the relevance classifier, its fusion with lexicon rules) are the weaker links and the right place to invest next.

7.2 Comparison with literature baselines and existing solutions

Compared with learned dialogue policies, the explicit policy sacrifices data-driven optimisation but gains deterministic reproducibility and direct inspection of decision paths. In a mature deployment with large annotated call corpora, a learned policy could be evaluated as an additional baseline. In the present low-resource setting, however, a learned approach would introduce a second major data requirement in addition to speech and entity recognition.

Compared with free-form generative assistants, the controlled realiser has intentionally limited language variation. This is not treated as a defect in the safety-critical context because every template can be reviewed and regression-tested. The measured increase in diversity shows that controlled templates can reduce repetition without abandoning bounded wording.

The comparison with the fixed-template baseline is a comparison with the group's earlier practice rather than with a published system. Its value is in showing that question variety and targeting can be improved on the same files. Relative to the POMDP tradition [8]-[10], the component gives up learned optimality in exchange for transparency and for the absence of a training requirement, which is appropriate for a domain with no dialogue data. Relative to generative assistants, it gives up fluency in exchange for auditability and bounded safety behaviour.

7.3 Research contribution

The contribution also includes a method of linking language-specific repair to protocol state. Sinhala numeral composition is not evaluated as an isolated NLP task; its value is demonstrated through the effect of a recovered house number on structured address completeness and subsequent policy targeting. This connection between lexical processing and dialogue readiness is central to the individual research component.

A second methodological contribution is the explicit treatment of off-topic handling as a finite policy with a safety veto. Many fallback systems are described only as classification plus a generic response. Here, the number of redirects, escalation behaviour and emergency override are represented as testable state transitions, making the behaviour suitable for regression evaluation.

The research contribution is threefold. First, a protocol-grounded clarification policy that combines dialogue state, slot completeness, distress and uncertainty in an explicit form, with per-field ask limits. Second, a composed numeral and structured address representation for spoken Sinhala locations, which separates house number, road and town and gives each a status. Third, a finite off-topic ladder with an emergency veto, evaluated on a multilingual gold set that includes adversarial and open-set items.

7.4 Product contribution

The product contribution is a clarification module that can be dropped into a call-taking pipeline. It returns a decision, a reason and a question for each turn, and it can be run and tested without any external service. The panel displays the strategy and the slot states so that an operator can see why a question was asked.

7.5 Limitations and threats to validity

Table 7.1 lists the principal limitations and the threats they pose. The most important are the size and authorship of the gold sets, the absence of a human-annotated clarification outcome, the weakness of the upstream entity model and the small number of speech-to-text calls checked. The author wrote the gold sets and designed the policy, so the results are not independent evidence; independent annotation would be needed to confirm them.

Threat	Effect on results	Mitigation or future work
Gold sets written by the author	Possible optimism in pass rates	Independent annotation of a held-out set
Small gold sets (8 to 146 items)	Wide uncertainty on each rate	Report counts; expand with panel logs
No human clarification outcome	Diversity is a proxy for quality	Expert rating of question appropriateness
Weak upstream entity model (F1 0.43)	Location and people recall limited	More labelled Sinhala and Tamil spans
Two speech-to-text calls compared	Recognition error not characterised	Larger reference set with WER and CER
Baseline is intentionally simple	Not a strong competitor	Compare with a rule-based slot-filler
Fusion override made routing worse, not better (Section 6.1)	Classifier override is not safe to enable by default	Calibrate or threshold the classifier before re-enabling fusion
Qualified road and town count as SUFFICIENT without a house number	Some calls may reach the dispatch-ready level without a house number	Decide the rule with call-takers; measure the effect on refine loops
Window sweep only shows the recall side (n=10)	Cannot see where a wider window would start accepting wrong numbers	Repeat the sweep on a larger, harder address gold set
Relevance fusion step not evaluated on its own	Deployed accuracy (77.5-86.7%) differs from the model-only benchmark (82.5%)	Ablate the lexicon fusion step the way the off-topic override was ablated

Table 7.5.1: Limitations and threats to validity

7.6 Implications

For practitioners, the results suggest that the most useful improvement to a call-taking assistant may lie in how it asks, not only in how well it recognises. For researchers, the structured location and the finite redirect ladder offer testable units that can be extended to other languages. For the emergency services, the component offers a way to make the questions asked by an automated line consistent with the protocol, provided that a human operator remains in the loop.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The value proposition is primarily operational rather than consumer-facing. The component aims to reduce avoidable repetition, preserve multilingual information and direct the caller toward missing protocol details. For a public emergency agency, potential value would be measured through reduced clarification turns, improved location completeness, consistent treatment of non-emergency calls and better operator visibility of why the system is asking a question.

Any future adoption should begin with an operator-assist scenario rather than autonomous call closure. This lowers deployment risk and allows call-takers to provide feedback on question appropriateness, local terminology and address conventions while the system remains advisory.

The primary users are the call-takers of disaster-management and emergency operations, who handle high volumes of calls with limited staff. The secondary users are callers in areas where the first language is Sinhala or Tamil and where automated lines are expected to improve response. The value proposition is a clarification assistant that asks better questions in the caller's language, captures house numbers spoken as words and keeps social calls out of the emergency queue.

8.2 Market and competitive position

The market for emergency call-handling software is served by large vendors whose products are closed and generally priced for national deployments. The component's position is narrower: it addresses the dialogue layer for low-resource languages, which such vendors have not prioritised. Its competitive advantage is transparency and language coverage, not scale. A commercial product would need integration with existing telephony and computer-aided dispatch systems, which the present prototype does not include.

8.3 Business or sustainability model

Sustainability depends more on governance and language maintenance than on compute cost. Lexicons and templates will require periodic review as terminology, place-name usage and operational procedures evolve. A viable maintenance model would therefore assign ownership of language resources and protocol configuration to a responsible technical team working with domain experts from the emergency agency.

The prototype's local execution supports predictable infrastructure cost and reduces reliance on external APIs, but production cost would also include telephony integration, secure hosting, monitoring, operator training, quality assurance and ongoing annotation. These costs should be considered in any future feasibility study rather than inferred from prototype runtime alone.

A sustainable model would be a public-sector licence with support, deployed inside the existing call-taking infrastructure of the operator. Revenue would come from integration and maintenance rather than from per-call fees, which keeps the service affordable for disaster management agencies. Running costs are low because the component is rule-based and runs locally; the main recurring cost is curation of the lexicons and templates by native speakers.

8.4 Scalability, intellectual property and compliance

Scalability has two dimensions. Technical scaling concerns concurrent calls and integration with production telephony; linguistic scaling concerns the addition of dialects, new address forms, code-switch patterns and reviewed templates. The current modular resource files support linguistic extension, but concurrency and high-availability behaviour were not evaluated in this thesis.

Compliance should include transparent documentation of decision-support scope, data minimisation, access control, retention, auditability and human oversight. Because the component operates in a public-safety context, deployment approval should be based on the responsible agency's legal and operational requirements rather than on prototype performance alone.

Scalability is mainly a question of adding languages and lexicons, which the design supports through resource files and the template structure. The lexicon and template content is the principal intellectual property and would need an open or licensed arrangement agreed with the group and the supervisors. Compliance requires that recorded calls are handled under the operator's data-protection rules, that caller data is minimised, and that the system remains a decision-support tool under human supervision.

8.5 Contribution to the Sustainable Development Goals

Table 8.1 maps the component to the Sustainable Development Goals. The contribution is indirect: it supports timely emergency response, which relates to health and safety, and it supports inclusive access for speakers of Sinhala and Tamil.

SDG	How the component contributes	Strength of link
SDG 3: Good health and well-being	Faster and more complete information in emergency calls	Indirect
SDG 11: Sustainable cities and communities	Better disaster response and preparedness	Indirect
SDG 10: Reduced inequalities	Service in the caller's own language	Direct
SDG 16: Peace, justice and strong institutions	Transparent, auditable decisions in public service	Indirect

Table 8.5.1: Contribution of the component to the Sustainable Development Goals

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

Across the study, the component demonstrates a coherent engineering principle: when data are limited and consequences are high, dialogue behaviour can be made safer and more testable by representing uncertainty explicitly and constraining the action space. The component does not solve every upstream recognition problem, but it provides a disciplined mechanism for recovering from partial information and for preventing uncontrolled conversational loops.

The results also show why end-to-end claims should be avoided. Improvements in question diversity and readiness classification are meaningful, yet dispatch-ready counts did not increase materially. The thesis therefore concludes with a validated component contribution and a clearly defined path toward stronger independent, user-centred and end-to-end evaluation.

This thesis designed, implemented and evaluated the dialogue-management component of an emergency call system for Sinhala, Tamil and English callers. The component replaces fixed templates with an explicit, protocol-grounded policy, recovers house numbers spoken as words, and governs off-topic callers with a finite ladder that cannot redirect a caller who has emergency evidence. On the author's gold sets and the group's call files, the component improved question diversity from 0.040 to 0.222 with a full safety pass rate, and it reduced the share of files classed NOT_READY_CRITICAL from 91.4 to 74.1 per cent, while the dispatch-ready share did not change.

9.2 Achievement of objectives

Objective	Status	Evidence
O1: Policy from state, readiness and affect	Achieved	Policy cases 8/8; comparison over 502 files
O2: Hundreds and structured address	Achieved (small gold)	Hundreds 8/8; address 10/10
O3: Finite off-topic ladder and veto	Achieved	Veto errors 0; no false INFO on 30 social items
O4: Acknowledging, component-specific questions	Achieved	Template matrix verified for three languages
O5: Gold evaluation and baseline comparison	Achieved	Results in Chapter 6; limitations in Section 7.5

Table 9.2.1: Achievement of the objectives in Section 1.5

9.3 Individual research and product contributions

The author's individual contributions are the extension of the numeral lexicon to hundreds, the structured address parser and its completeness rule, the ask-limit and road-first changes in the policy, the controlled templates for refine questions, and the evaluation scripts and gold sets for these parts. The dialogue state, the wider policy and the off-topic and affect modules existed before these changes and were extended by the author; the thesis distinguishes the two where it matters, and the author confirms that the description of contribution reflects the work done.

9.4 Future directions

Future evaluation should prioritise independent evidence. A held-out set labelled by more than one annotator would allow agreement to be measured, while call-taker review could score relevance, clarity, tone and operational usefulness of each clarification. An end-to-end test set with reference transcripts and protocol labels would then allow speech-recognition, entity-extraction and policy errors to be separated on the same calls.

Future technical work should preserve the current safety invariants while testing stronger alternatives. Candidate extensions include confidence-calibrated slot status, a learned ranking model constrained by protocol rules, language-specific address resources, and richer contradiction handling across turns. Any generative component should remain downstream of a bounded decision and should be validated against the same safety and regression suite before use.

- Run the planned ablation (A0 baseline, A1 rules only, A2 classifier only, A3 full fusion) and report each condition's false information rate.
- Obtain an independent annotation of a held-out clarification set and an expert rating of question appropriateness.
- Extend the speech-to-text evaluation to a larger reference set with word and character error rates by language.
- Separate the resource lexicon used for hazard detection from the one used for information requests, to correct the banding errors found in triage.
- Add a confirmation step for the full structured address before dispatch, and test its effect on repeated turns.
- Evaluate the component in a supervised live pilot with call-takers, measuring time to a dispatch-ready location.

REFERENCES

[1] A. Conneau et al., "Unsupervised cross-lingual representation learning at scale," in Proc. 58th Annu. Meeting Assoc. Comput. Linguistics (ACL), 2020, pp. 8440-8451.

[2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in Proc. NAACL-HLT, 2019, pp. 4171-4186.

[3] A. Vaswani et al., "Attention is all you need," in Advances in Neural Information Processing Systems (NeurIPS), 2017.

[4] G. Lample, M. Ballesteros, S. Subramanian, K. Kawakami, and C. Dyer, "Neural architectures for named entity recognition," in Proc. NAACL-HLT, 2016, pp. 260-270.

[5] Z. Huang, W. Xu, and K. Yu, "Bidirectional LSTM-CRF models for sequence tagging," arXiv:1508.01991, 2015.

[6] J. Lafferty, A. McCallum, and F. Pereira, "Conditional random fields: Probabilistic models for segmenting and labeling sequence data," in Proc. 18th Int. Conf. Mach. Learn. (ICML), 2001, pp. 282-289.

[7] D. Nadeau and S. Sekine, "A survey of named entity recognition and classification," Lingvisticae Investigationes, vol. 30, no. 1, pp. 3-26, 2007.

[8] J. D. Williams, A. Raux, D. Ramachandran, and A. Black, "The dialog state tracking challenge," in Proc. SIGDIAL, 2013, pp. 404-413.

[9] J. D. Williams and S. Young, "Partially observable Markov decision processes for spoken dialog systems," Computer Speech & Language, vol. 21, no. 2, pp. 393-422, 2007.

[10] S. Young, M. Gasic, B. Thomson, and J. D. Williams, "POMDP-based statistical spoken dialog systems: A review," Proc. IEEE, vol. 101, no. 5, pp. 1160-1179, 2013.

[11] O. Lemon and O. Pietquin, Eds., Data-Driven Methods for Adaptive Spoken Dialogue Systems. New York, NY, USA: Springer, 2012.

[12] M. Wooldridge and N. R. Jennings, "Intelligent agents: Theory and practice," The Knowledge Engineering Review, vol. 10, no. 2, pp. 115-152, 1995.

[13] OASIS, "Common Alerting Protocol Version 1.2," OASIS Standard, 2010.

[14] T. Wolf et al., "Transformers: State-of-the-art natural language processing," in Proc. EMNLP: System Demonstrations, 2020, pp. 38-45.

[15] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," J. Mach. Learn. Res., vol. 12, pp. 2825-2830, 2011.

[16] European Emergency Number Association (EENA), emergency call-taking operational guidance (Where, What, Who, When call-taking categories). [Author to confirm exact title, edition and year before submission.]

[17] W. J. Scheirer, A. de Rezende Rocha, A. Sapkota, and T. E. Boult, "Toward open set recognition," IEEE Trans. Pattern Anal. Mach. Intell., vol. 35, no. 7, pp. 1757-1772, 2013.

[18] S. Larson et al., "An evaluation dataset for intent classification and out-of-scope prediction," in Proc. EMNLP-IJCNLP, 2019, pp. 1311-1316.

[19] P. Joshi, S. Santy, A. Budhiraja, K. Bali, and M. Choudhury, "The state and fate of linguistic diversity and inclusion in the NLP world," in Proc. 58th Annu. Meeting Assoc. Comput. Linguistics (ACL), 2020, pp. 6282-6293.

[20] N. de Silva, "Survey on publicly available Sinhala natural language processing tools and research," arXiv:1906.02358, 2019.

[21] J. L. Austin, How to Do Things with Words. Oxford, U.K.: Oxford Univ. Press, 1962.

[22] J. Allen and M. Core, "Draft of DAMSL: Dialog act markup in several layers," Multiparty Discourse Group, Discourse Research Initiative, 1997.

[23] M. Henderson, B. Thomson, and J. Williams, "The second dialog state tracking challenge," in Proc. SIGDIAL, 2014, pp. 263-272.

[24] P. Budzianowski et al., "MultiWOZ: A large-scale multi-domain Wizard-of-Oz dataset for task-oriented dialogue modelling," in Proc. EMNLP, 2018, pp. 5016-5026.

[25] A. Conneau and G. Lample, "Cross-lingual language model pretraining," in Advances in Neural Information Processing Systems (NeurIPS), vol. 32, 2019.

GLOSSARY

Clarification: A system turn that asks the caller for a missing or uncertain piece of protocol information.

Dispatch readiness: A computed level (DISPATCH_READY, CLARIFICATION_REQUIRED, NOT_READY_CRITICAL) that states whether a call holds enough information to send responders.

Hedge: A caller utterance that signals uncertainty, for example "I think" or "maybe", which lowers the confidence of the slot it fills.

Numeral composer: A lexicon-driven component that combines number words (tens, ones, hundreds) into a single value, for example 625.

Protocol slot: A field of the emergency protocol: WHERE (location), WHAT (incident), WHO (people), LIFE (injury or entrapment).

Redirect ladder: The finite sequence of off-topic responses (soft, firm, final) that ends the call if the caller remains off-topic.

Structured address: A decomposition of a location into house number, road and town, each with its own source and status.

Code-switching: Speech that mixes two languages within a sentence, common in Sri Lankan calls (for example, Sinhala with English road names).

Controlled template: A question sentence with placeholders filled from the dialogue state, used instead of free generation.

Emergency veto: The rule that an off-topic route is never assigned when emergency evidence is present in the turn.

Gold set: A set of utterances with labels written before evaluation, used to measure the accuracy of a component.

Lexicon: A resource file of word forms and their values, read by the numeral composer and the triage module.

Span: A contiguous part of the transcript that a model or rule assigns to a field, such as an address or a count.

Speech-to-text (STT): The service that converts recorded speech into the transcript used by the component.

Slot status: The completeness state of one protocol field: SUFFICIENT, WEAK, MISSING or UNTRUSTED.

Question diversity: The number of distinct questions divided by the number generated; a higher value means less repetition.

APPENDICES

Appendix A: Evaluation datasets and reproduction commands

The evaluation scripts are in the backend folder of the component. Each command prints the score and writes a JSON result file under results/ (or research/off_topic_control/results/). The scripts are deterministic on the gold sets provided.

python eval_numeral_composer.py (numeral composer, 19 gold items, address guard, 5 items)

python eval_structured_address.py (structured address, 10 items; hundreds of numerals, 8 items)

python eval_adaptive_call_triage.py (policy cases, 8 items; triage gold, 20 items)

python eval_window_sensitivity.py (house-number evidence window sweep, structured address gold)

python eval_relevance_calibration.py (relevance classifier reliability on dev.jsonl and test.jsonl)

The off-topic routing evaluation (146 items), its ablation (eval_ablation_off_topic.py) and the clarification comparison over the 502 call files are produced by the scripts under research/off_topic_control/ and compare_clarification_research.py.

9.4.1 Error analysis

Of the 146 off-topic items, 14 were routed incorrectly. Table A.1 lists them. The errors fall into four groups, and each group points to a specific cause rather than to random noise.

Item	Bucket	Gold	Predicted	Difficulty	Language
ot_os_005	open set holdout	OFF_TOPIC	HIGH_EMERGENCY	hard	si-en
ot_wx_002	weather chat	OFF_TOPIC	INFO	hard	si-LK
ot_wx_003	weather chat	OFF_TOPIC	INFO	hard	ta-LK
ot_wx_005	weather chat	OFF_TOPIC	HIGH_EMERGENCY	hard	si-en
info_013	status inquiry	INFO	OFF_TOPIC	easy	en
info_018	status inquiry	INFO	OFF_TOPIC	easy	si-en
info_019	status inquiry	INFO	OFF_TOPIC	easy	en
svc_010	welfare resource	SERVICE	OFF_TOPIC	easy	ta-LK
svc_012	welfare resource	SERVICE	OFF_TOPIC	easy	si-LK
med_008	external hazard	MEDIUM	OFF_TOPIC	easy	si-LK

Item	Bucket	Gold	Predicted	Difficulty	Language
adv_001	ambiguous borderline	OFF_TOPIC	INFO	adversarial	en
adv_004	ambiguous borderline	SERVICE	HIGH_EMERGENCY	hard	si-LK
adv_007	ambiguous borderline	OFF_TOPIC	HIGH_EMERGENCY	hard	si-LK
adv_008	ambiguous borderline	OFF_TOPIC	HIGH_EMERGENCY	easy	en

Table 9.4.1: Off-topic routing errors (14 of 146)

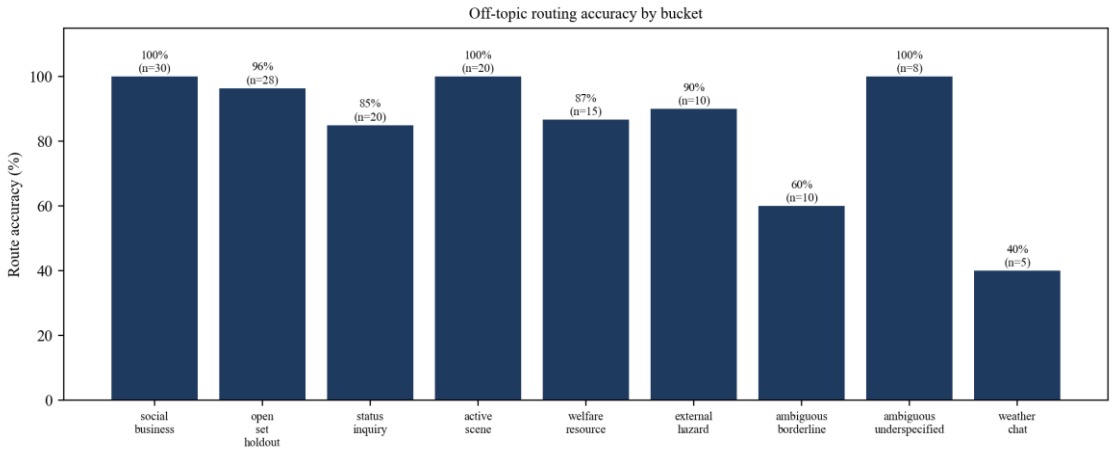


Figure A.1: Off-topic routing accuracy by bucket

- Group 1, hazard vocabulary without life risk (ot_os_005, ot_wx_005): a burned bulb, and a traffic problem caused by rain, were predicted as HIGH_EMERGENCY because a hazard word was present, and the absence of a life-risk statement was not used.
- Group 2, weather description treated as information (ot_wx_002, ot_wx_003): weather talk was routed to INFO. The information cue is the weather word itself, which is too broad.
- Group 3, status questions treated as off-topic (info_013, info_018, info_019): direct status questions about school closures, a flood warning and a landslide were routed to OFF_TOPIC. These are short questions with no emergency words, and the INFO gate depends on emergency words.
- Group 4, resource and hazard requests treated as off-topic (svc_010, svc_012, med_008), and borderline ambiguous items (adv_001, adv_004, adv_007, adv_008). The resource requests share their cause with the triage misses in Table 6.8.

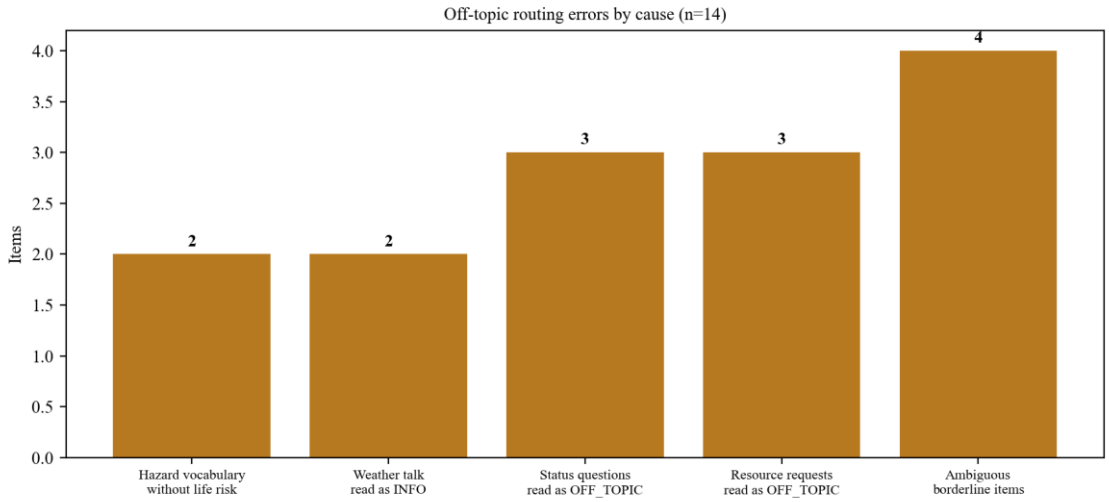


Figure A.2: Off-topic routing errors grouped by cause

The triage gold set has two misses, both requests for drinking water and food at a shelter. Table A.2 gives the two items. The route was correct, DIRECT_INFORMATION_RESPONSE, and the band was MEDIUM rather than INFO. The resource request contains cues that the hazard detector also uses, so the banding treats it as a hazard.

Utterance	Expected band	Predicted band	Route
We need drinking water and food at the shelter	INFO	MEDIUM	DIRECT_INFORMATION_RESPONSE (correct)
ඊයේ වැස්ස නිසා අපි ඔක්කොම පන්සලට ආවා දැන් අපි කාටවත් වතුර බොන්න නැහැ පුළුවන් නම් අපිට වතුර දෙන්න පුළුවන්ද	INFO	MEDIUM	DIRECT_INFORMATION_RESPONSE (correct)

Table A.2: Triage gold items that were banded incorrectly

Taken together, the error groups suggest three changes. The resource lexicon should be separated from the hazard lexicon, so that a request for water is not evidence of a hazard. Status questions should be recognised by their interrogative form, not only by emergency words. The hazard cue should be checked for a life-risk statement before the call is banded as a hazard. These changes are listed as future work in Section 9.4 and have not been implemented, so the results in this chapter describe the component as it stands.

9.4.2 Worked example of one call

Table A.3 traces one call through the component, using outputs recorded during development. The caller reports a fire at home in Nuwara Eliya, then gives the full address. The table shows how the location moves from WEAK to SUFFICIENT and how the policy then moves to the people question.

Turn	Caller utterance (abridged)	Location status	Policy and question
1	අපේ ගෙදර ගිනි ගන්නවා අපි ඉන්නේ නුවර එළියේ	WEAK (town only; no road, no house number)	ASK_REFINE for the road, acknowledging the town
2	මගේ ඇඩ්ඩස් එක හයසිය විසිපහ නුවර එළියපාර ගම්පල	SUFFICIENT (house 625, road, town); exact span 625 නුවර එළියපාර ගම්පල	Location complete; ASK_DIRECT for the number of people

Table A.3: Worked example: location status and next question across two turns

The example also shows a limit of the present rule. A town without a road is WEAK, so the question returns to location. A road and a town without a house number, however, are still classed SUFFICIENT by the existing qualified-area rule, which was left unchanged to avoid repeated refine loops. Whether a house number should be required is an open design decision, discussed in Section 7.5.

9.4.3 Ask-limit simulation and question variants

Table A.4 shows the effect of the ask limit on the target that the policy selects. The simulation sets the number of earlier location asks and asks the real selection function for the next target. Under the earlier limit of two, the policy advanced to the people question after two location asks. Under the current limit of four, it holds the location for up to three asks and advances only at four.

Earlier location asks	Next target	Status
0	incident_location	WEAK
1	incident_location	WEAK
2	incident_location	WEAK
3	incident_location	WEAK
4	people_count	MISSING

Table A.4: Next target by number of earlier location asks

Table A.5 lists the question variants that the controlled templates produce for a caller who has named the town of Nuwara Eliya. Each language has an acknowledgement variant and a retry variant, and the table is generated from the template function so that it matches the code exactly.

Language	Missing part	First ask	Retry
si-LK	road	ඔයා නුවර එළියේ කිව්වා. ඒ ප්‍රදේශයේ ඔයා ඉන්න පාරේ නම කියන්න පුළුවන්ද?	පාරේ නම හරියටම දන්නේ නැත්නම්, ළඟම තියෙන ප්‍රධාන පාර, කඩයක් හෝ පන්සලක් වගේ දෙයක් කියන්න.
si-LK	house no	ඔයා නුවර එළියේ කිව්වා. ගෙදර අංකය හෝ ගෙදරේ නම කියන්න පුළුවන්ද?	ගෙදර අංකය හරියටම දන්නේ නැත්නම්, ගෙදර ළඟ තියෙන කඩයක් හෝ ලොකු ස්ථානයක් කියන්න.
en	road	You said නුවර එළියේ. What is the name of the road you are on there?	If you don't know the road name, tell me the nearest main road or a landmark such as a shop or temple.
en	house no	You said නුවර එළියේ. What is the house number or house name?	If you don't know the number, tell me a nearby shop or landmark next to the house.
ta-LK	road	நீங்கள் நுவர எළியේ என்றீர்கள். நீங்கள் அங்கே இருக்கும் தெருவின் பெயர் சொல்ல முடியுமா?	தெருவின் பெயர் தெரியாவிட்டால், அருகில் உள்ள பெரிய சாலை அல்லது கடை, கோவில் போன்ற அடையாளம் சொல்லுங்க.
ta-LK	house no	நீங்கள் நுவர எළியේ என்றீர்கள். உங்கள் வீட்டு எண் அல்லது வீட்டின் பெயர் சொல்ல முடியுமா?	வீட்டு எண் தெரியாவிட்டால், வீட்டின் அருகில் உள்ள கடை அல்லது அடையாளம் சொல்லுங்க.

Table A.5: Address question variants by language and missing component

9.4.4 Ablation study

Section 6.1's off-topic result is produced by one condition of a four-way ablation that the design in Section 4.4 anticipated. Table A.6 reports all four: A0, no off-topic gate at all, so every item is treated as an ongoing emergency; A1, the lexicon and triage rules alone, which is the function behind the headline result; A2, the fine-tuned relevance classifier alone, with no rules; and A3, the rules with a classifier override to OFF_TOPIC and an emergency veto, which was the fusion design intended for the component.

Condition	Route accuracy	OFF_TOPIC F1	False INFO on social	Emergency veto errors
A0, no gate	0.144	0.000	0/30	0
A1, rules only (headline)	0.904	0.904	0/30	0
A2, classifier only	0.514	0.743	0/30	1
A3, rules plus classifier fusion	0.801	0.816	0/30	0

Table A.6: Ablation of the off-topic router (146 items)

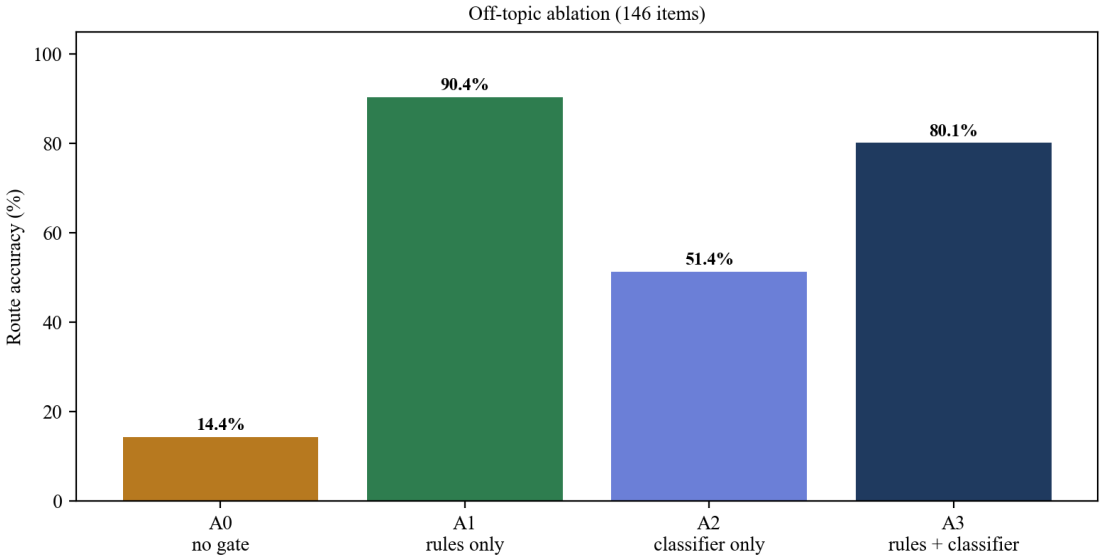


Figure A.3: Off-topic ablation: route accuracy by condition

The rules-only condition (A1) is the strongest of the four, and it is what Table 6.1's headline result already reports. The fusion condition (A3) is weaker than rules alone, 80.1% against 90.4%, because the classifier's own errors are injected into the routing decision through the override. The classifier-only condition (A2) reaches 51.4%, which is expected: it has four coarse relevance labels and no class for MEDIUM, INFO, SERVICE or AMBIGUOUS, so it cannot be correct on those items by construction. The no-gate condition (A0) reaches 14.4%, which is the share of the gold set that is genuinely HIGH_EMERGENCY, and it shows the value of having a gate at all, even a simple one.

The ablation also explains a gap between the design in Chapter 4 and the headline result. The script that produces the headline off-topic numbers calls a function named predict_relevance in the relevance module to compute a classifier override, but no

function of that name exists there, only the RelevancePredictor class does. The import fails, a bare exception handler hides the failure, and the override is silently never applied. The headline result is therefore a rules-only result, not the fusion result the design intended, and by this ablation it is also the more accurate of the two. The fault is confined to this evaluation script: services/live_emergency_pipeline.py calls the RelevancePredictor class directly for other purposes and was not affected.

9.4.5 Parameter sensitivity and calibration

Two further checks go beyond pass/fail accuracy: whether a chosen parameter was arbitrary, and whether a model's confidence can be trusted. Both follow a pattern used for retrieval and matching systems, where a candidate-window size is swept against recall, and a classifier's predicted probability is checked against its observed correctness on held-out data; the same pattern applies here, to a structural parameter and to a learned classifier respectively.

9.4.5.1 House-number evidence window

The structured address parser accepts a numeral as a house number only when an address cue word is within a fixed number of tokens before it, or a road word is within a fixed number of tokens after it (Section 4.4, structured address). The deployed values are 3 tokens before and 2 after. Figure A.3 sweeps each independently against the 10-item structured address gold set, holding the other fixed at its deployed value.

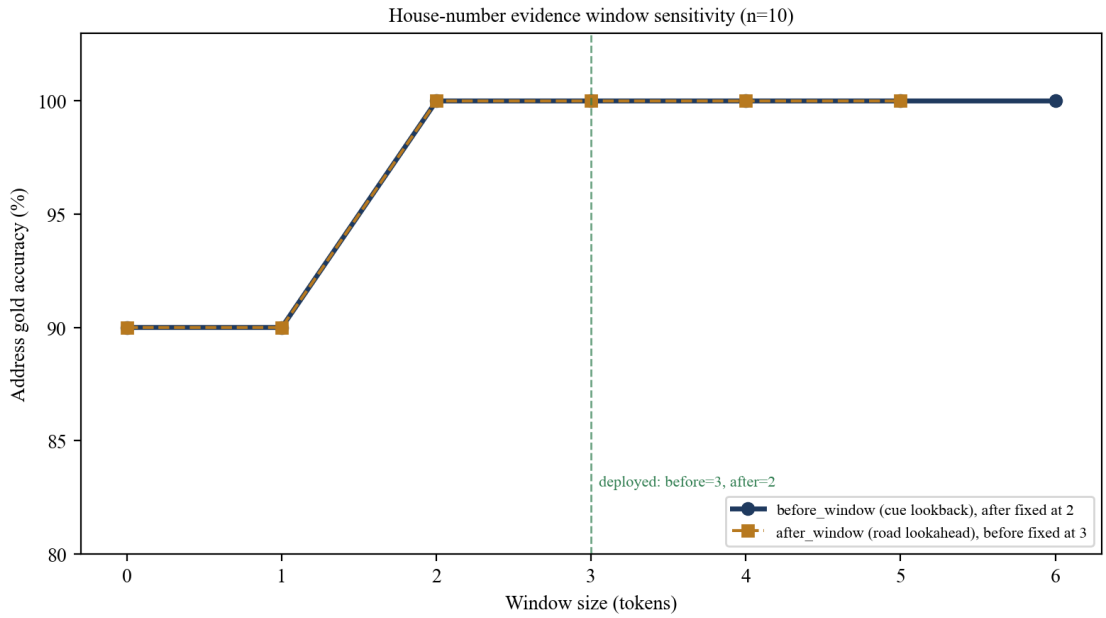


Figure A.4: House-number evidence window sensitivity

Window	Size (tokens)	Correct / n	Accuracy
before (cue lookback)	0	9/10	90%
before (cue lookback)	1	9/10	90%
before (cue lookback)	2	10/10	100%
before (cue lookback)	3	10/10	100%
before (cue lookback)	4	10/10	100%
before (cue lookback)	5	10/10	100%
before (cue lookback)	6	10/10	100%
after (road lookahead)	0	9/10	90%
after (road lookahead)	1	9/10	90%
after (road lookahead)	2	10/10	100%
after (road lookahead)	3	10/10	100%
after (road lookahead)	4	10/10	100%
after (road lookahead)	5	10/10	100%

Table A.7: House-number evidence window sweep

Accuracy rises from 90% at a window of 0-1 tokens to 100% once the window reaches 2 tokens, and stays at 100% up to 6 tokens for both the cue-lookback and the road-lookahead window. This shows the deployed window has margin: it is not sitting at the edge of a cliff where one fewer token would lose a real house number. It does not show the other side of the trade-off, because the gold set (10 items) contains no case where a wider window would wrongly accept a number that is not a house number; a larger gold set would be needed to find the point where widening the window starts hurting precision rather than helping recall.

9.4.5.2 Relevance classifier calibration

The relevance classifier reports a confidence with every prediction. Figure A.4 checks whether that confidence is trustworthy: for callers the model says it is p confident about, is it actually correct about p of the time? The check uses the model's own held-out splits, dev.jsonl (30 items) and test.jsonl (40 items), from data/relevance_research/, the same split behind the headline relevance metrics in Table 6.5, so there is no leakage from training.

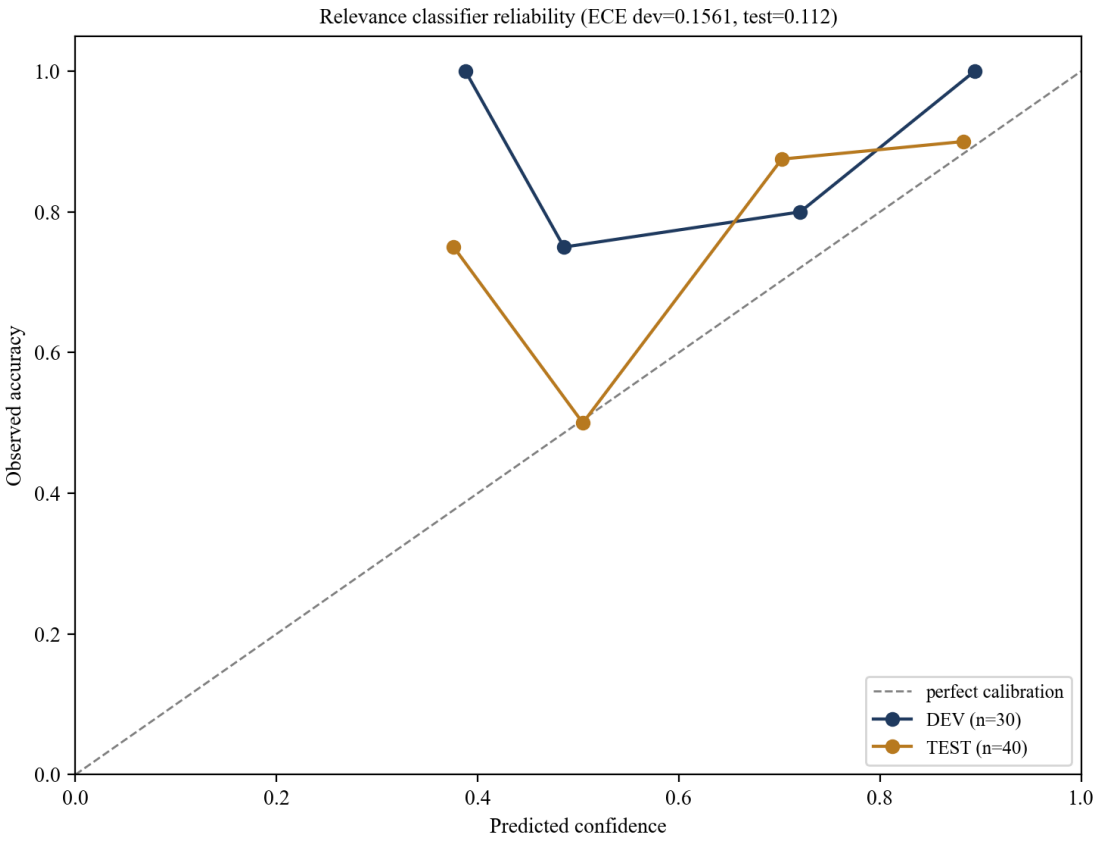


Figure A5: Relevance classifier reliability: DEV and TEST

Split	Confidence bin	Mean confidence	Observed accuracy	n
DEV	0.2-0.4	0.388	100%	1
DEV	0.4-0.6	0.486	75%	8
DEV	0.6-0.8	0.721	80%	10
DEV	0.8-1.0	0.894	100%	11
TEST	0.2-0.4	0.376	75%	4
TEST	0.4-0.6	0.505	50%	10
TEST	0.6-0.8	0.702	88%	16
TEST	0.8-1.0	0.883	90%	10

Table A.8: Relevance classifier reliability bins

Expected calibration error is 0.1561 on DEV and 0.112 on TEST: on average, the model's stated confidence is off by about 11-16 percentage points from what actually happens, which is a moderate miscalibration rather than a severe one. The curves are also noisy in the middle confidence range, which is expected with bins as small as a handful of items each. A second finding here is a gap worth naming: deployed accuracy through the full predict() pipeline is 86.7% on DEV and 77.5% on TEST, lower than the test accuracy of 82.5% reported for the model in isolation in Table 6.5. The difference is that Table 6.5 scores the raw model output, while this calibration check scores RelevancePredictor.predict(), which applies an auxiliary lexicon-based fusion step on top of the model before returning a label (Section 4.4). That fusion step is not yet itself evaluated separately, which is a gap this check surfaced rather than one it was designed to find.

Appendix B: Academic integrity and AI use disclosure

Tools used: Claude (Anthropic), via the Claude desktop application and Claude Code, a command-line coding assistant.

Purpose and sections affected: (i) code assistance for the numeral composer extension, the structured address parser, the clarification policy adjustment and the controlled template realiser; (ii) drafting support for the structure and wording of this thesis; (iii) preparing the figures from repository result files.

Verification: every code change was tested with the gold scripts listed in Appendix A before it was accepted, and every number in Chapter 6 was read from the result files rather than typed. The author reviewed and remains responsible for all content, analyses and conclusions.